



COMSATS University Islamabad (CUI)

**Software Requirement Specification
(SRS DOCUMENT)
for**

SkillEdge AI

Version 1.0

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Bachelor of Science in Artificial Intelligence (2022-2026)

Revision History

Name	Date	Reason for Changes	Version

Application Evaluation History

Comments (by committee) *include the ones given at scope time both in doc and presentation	Action Taken
Scope: Overall ok, but need to be clear on sequence of modules	We appreciate the panels input about our project. We have adjusted the modules accordingly.
Scope: How they will generate new questions after training?	We will Use generative models like T5 to generate new questions each time.

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Signature_____

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1. Introduction

SkillEdge AI, a web-based platform for interview, is guided by this Software Requirements Specification (SRS) document, offering a comprehensive overview of its development and functionality. The primary purpose of this document is to articulate the detailed requirements, features, and constraints that will govern the design and implementation of SkillEdge AI. As an innovative solution, SkillEdge AI aims to revolutionize the way candidates prepare for interviews, providing a dynamic and interactive environment for skill enhancement.

1.1 Purpose

SkillEdge AI is a cutting-edge web-based tool that aims to transform interview preparation using AI-driven simulations and instant feedback. For students, job applicants, or professionals looking to sharpen their communication and technical skills, SkillEdge AI offers an in-depth and engaging learning experience.

The tool mimics real-life interview situations across various fields, leveraging sophisticated Natural Language Processing (NLP) and machine learning algorithms. From specialized Technical Interviews featuring AI-based answer analysis to Behavioral Assessments evaluating speech pronunciation, timing, and self-assurance, SkillEdge AI prepares users adequately for actual interview situations. Going beyond the limitations of traditional mock interviews, SkillEdge AI adds modules for Performance Reports with elaborate analytics, Interactive Chatbots offering individualized mentorship, and Progress Tracking that helps track improvements over time.

By providing an intelligent, adaptive, and user-focused method of interview training, SkillEdge AI enables users to gain confidence, improve their communication skills, and become job market competitive. The aim is to establish a dynamic, AI-based environment where users are provided with actionable insights and ongoing improvement opportunities, ultimately leading to career success and professional development.

1.2 Scope

The scope of SkillEdge AI is developing the AI-powered interview preparation platform, which would ensure users a lifelike, dynamic, and tailored mock interview experience. The platform would implement mock interview scenarios in any number of professional domains with domain specific questions pertinent to the profession that the user is interested in. Answers from the user will receive in-depth, AI-generated feedback, with two elements: first, verbal content, meaning answer relevance, correctness, structure, and completeness, and second, non-verbal elements like voice clarity, pace, and filler words.

It will also wrap up with a thorough performance report detailing strengths and areas to improve upon and feature an interactive chatbot that steps the user through contextual advice on how to

get better. The users of SkillEdge AI will also be able to track their progress over a few sessions, giving them a means of viewing improvement over time. Full access on the platform will be given via a very user-friendly web interface, so that navigation is seamless, and engagement promotes regular practice.

In addition, SkillEdge AI will support various customizable interviewer personas to enable simulation of various interview styles and adaptive question based on user performance. The users can get their answers compared to ideal ones for more accurate learning. SkillEdge AI offers a rich combination of content-based and delivery-focused feedback and will be truly comprehensive and serve as an all-in-one tool for interview preparation. Users will be empowered to approach interviews confidently, clearly, and professionally.

1.3 Modules

1.3.1 Module 1: User Registration

The User Login and Onboarding module enables users to create accounts, log in, and provide essential information that personalizes their interview preparation experience. This initial setup captures details like the user's area of expertise, profession, and interview goals.

***FE-1:** New users sign up by providing their name, email, password, and information such as their area of expertise, preferred domains (e.g. technical, HR), desired roles, whether they are a student or professional, and their current career goals.*

***FE-2:** Users log in using their username and password to access their personalized interview dashboard.*

***FE-3:** Users can specify their interview goals, such as preparing for a particular type of job or skill, to tailor the interview experience based on their objectives.*

1.3.2 Module 2: Interview Simulation

The Interview Simulation module will provide users with a real interview scenario, in which there will be an avatar (person) which will be asking questions, and the user will answer them.

***FE-1:** The system generates a set of domain-specific questions tailored to the user's expertise and experience level.*

***FE-2:** An avatar asks questions from the user, simulating a real interview environment to create a more engaging experience.*

***FE-3:** A certain time limit will be provided for complex questions, helping users practice thinking and responding quickly under pressure.*

1.3.3 Module 3: Interview from Resume

The CV-Based Interview module allows candidates to upload their CVs, which the system then analyzes to pull out important details and then take the interview.

FE-1: Users will be able to upload their CV.

FE-2: System will extract important key points from cv like skills, education, experience etc.

FE-3: According to the extracted information, the system will ask questions from the user, questions will address his technical, behavioral and domain-specific knowledge and show him the corresponding results.

1.3.4 Module 4: Evaluate Verbal/Non-Verbal Response

The Non-Verbal Feedback module assesses various non-verbal aspects of communication, helping users improve delivery style and overall presence.

FE-1: User responses are compared to model answers to check whether they are correct or not and then highlights answers which are not correct.

FE-1: The system analyzes the user's speech rate, providing feedback on whether they are speaking too quickly or too slowly.

FE-2: The system detects filler words (e.g., "um," "uh") and provides feedback on clarity and articulation.

FE-4: Feedback on tone and pitch is provided, helping users improve their vocal delivery to sound more confident and engaging.

1.3.5 Module 5: Interview Highlights

The Performance Report module generates a comprehensive report after each interview session, highlighting key metrics and offering targeted feedback.

FE-1: User can access 2 separate reports representing verbal and non-verbal responses.

FE-2: User can see his performance from details, graphs, bars provided in the reports.

FE-3: Performance is broken down into categories like technical knowledge and communication skills.

FE-4: At the end of each interview, there will be a detailed feedback section explaining why certain responses were strong or how could be improved.

1.3.6 Module 6: Overall Feedback

The Answer Analysis and Feedback module evaluates the user's responses for accuracy, relevance, structure, and completeness, using AI-driven content analysis.

FE-1: The system evaluates each answer for correctness, relevance, completeness and domain- specific accuracy.

FE-2: Feedback is provided on the structure of the response, including sound clarity, and completeness.

FE-3: User responses are compared to model answers to check whether they are correct or not and then highlights answers which are not correct.

FE-4: A score is generated for each answer, covering areas like relevance, depth, and alignment with the question, helping users understand their performance level. This score is used at the end to compute overall performance

1.3.7 Module 7: Interactive Chatbot

The Interactive Chatbot module provides personalized guidance by allowing users to engage in a follow-up discussion based on their performance report.

FE-1: Users can ask the chatbot questions about specific feedback points to gain a deeper understanding of their strengths and weaknesses.

FE-2: The chatbot suggests resources (e.g., articles, videos) and improvement tips tailored to the user's needs.

FE-3: Based on the user's interview goals, the chatbot recommends specific actions to help them progress toward those objectives.

1.3.8 Module 8: Progress Tracking and Analytics

The Progress Tracking and Analytics module provides users with insights into their improvement over time and helps them monitor their development across various skills.

FE-1: Users can review past performance reports to observe trends in their improvement over multiple sessions.

FE-2: The system shows trends in skills like communication, technical knowledge, and non-verbal communication, allowing users to understand which areas are improving.

FE-3: Users can set interview preparation goals and track their progress toward achieving them, based on metrics like improved scores and reduced filler word usage.

FE-4: Progress data is displayed on a user-friendly dashboard with visualizations like graphs and charts, making it easy to interpret development over time.

1.4 Overview

This document provides detailed specifications and functionalities of the SkillEdge AI application. For the overview of the project, use case and the system context diagram is presented to provide a graphical representation of user and system perspective respectively. Moreover, comprehensive details are provided via detailed use cases, functional requirements and an event response table. The document further explains non-functional requirements and external interface requirements.

2. Overall Description

2.1 Product Perspective

SkillEdge AI is a groundbreaking, standalone web application designed to transform the interview and language proficiency training landscape. Unlike traditional interview preparation tools that rely on static question banks and one-dimensional feedback, SkillEdge AI delivers a fully interactive, dynamic simulation environment that combines both technical and behavioral interview practices. It leverages advanced AI and machine learning techniques to generate domain-specific questions in real time and evaluate user responses with a focus on both content accuracy and delivery. SkillEdge AI not only helps users polish their technical and behavioral skills but also provides the confidence required to perform well in real-life interviews, which makes it an invaluable career advancement tool.

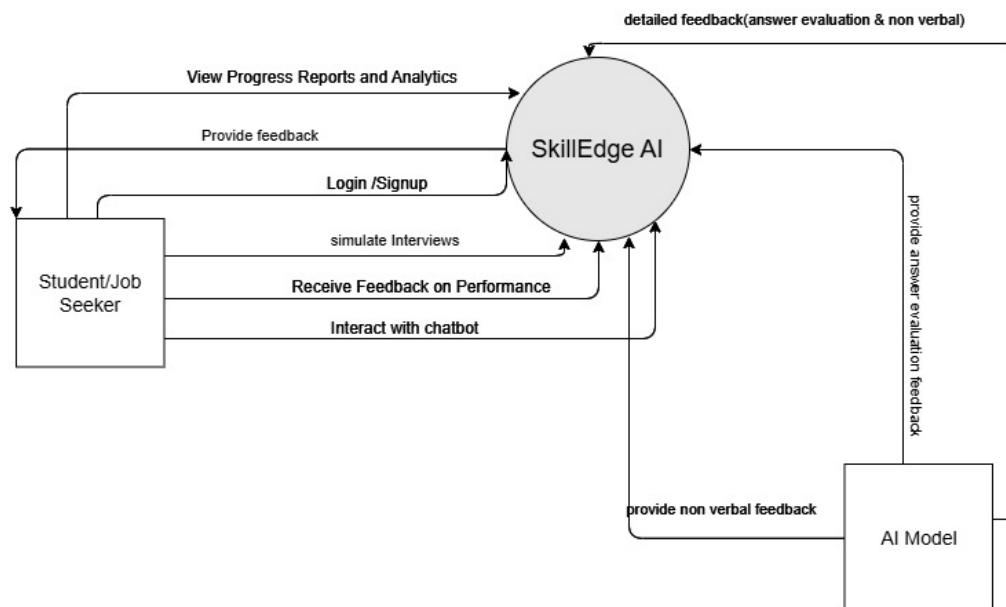


Figure 1: Context Diagram of the SkillEdge AI

2.2 User classes and characteristics

Table 1 Shows user classes and characteristics for SkillEdge AI

User Classes	Description
Students/Job Seekers	Individuals preparing for job interviews, ranging from university students and recent graduates to experienced professionals seeking career advancement or a job change. They use SkillEdge AI to practice both behavioral and technical interviews, receive immediate and personalized feedback on communication skills (including verbal aspects and non-verbal audio cues), and track their progress over multiple sessions. Their needs include realistic simulations, adaptive question generation, detailed performance reports, and actionable insights for improvement.

2.3 Operating Environment

OE-1: The SkillEdge AI s application shall be compatible with the following operating systems: Windows 10 and later versions and macOS 10.13 and later versions.

OE-2: The application requires a minimum hardware configuration of 2 GHz dual-core processor, 4 GB RAM.

OE-3: Users shall access the SkillEdge AI application through the following web browsers: Google Chrome (latest version), Mozilla Firefox (latest version), Microsoft Edge (latest version) and Safari (latest version).

OE-4: The application shall integrate with popular email services for user registration and communication, including Gmail and Google.

OE-5: The SkillEdge AI application shall comply with international data protection regulations ensuring the secure storage and processing of user data.

2.4 Design and Implementation Constraints

CON-1: SkillEdge AI is constrained to a Python-based technology stack, utilizing Fast API for backend development to leverage its asynchronous capabilities and efficient handling of concurrent requests.

CON-2: The frontend development is constrained to React.js, chosen for its ability to build dynamic, responsive, and user-friendly interfaces that integrate seamlessly with the Fast API backend.

CON-3: The backend development is constrained to Fast API, which provides a lightweight, high-performance, and event-driven architecture ideal for building modern RESTful APIs and real-time applications in Python.

CON-4: MongoDB is the chosen database engine, integrated via an asynchronous Python library (such as Motor), providing a flexible NoSQL solution for managing diverse data types and supporting scalability.

CON 5: SkillEdge AI integrates WebSocket support using Fast Api's built-in capabilities to enable real-time communication, ensuring instantaneous and seamless interactions between users and the application.

3. Requirement Identifying Technique

3.1 Use Case Diagram for User

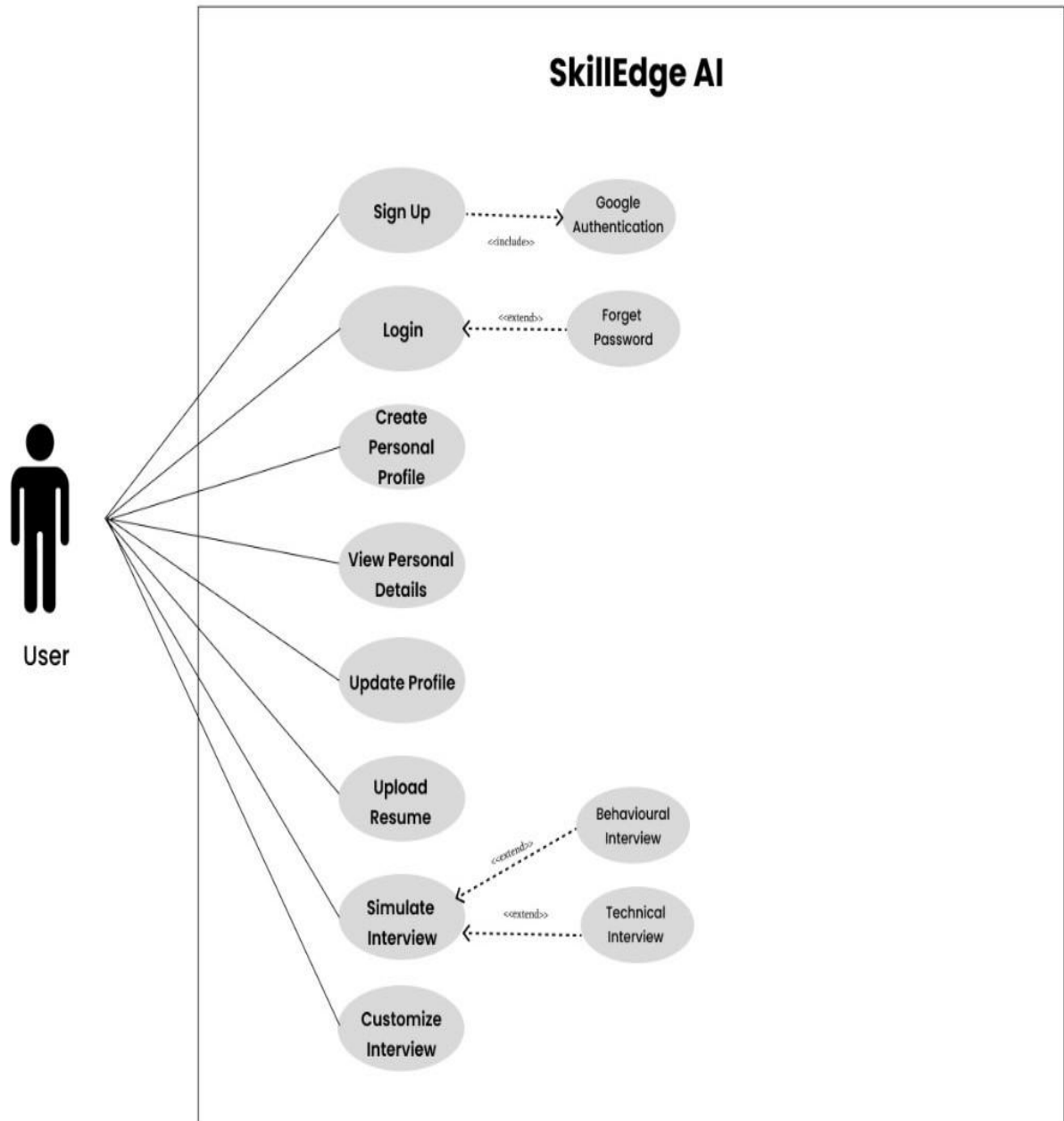


Figure 2: Use Case Diagram (1)

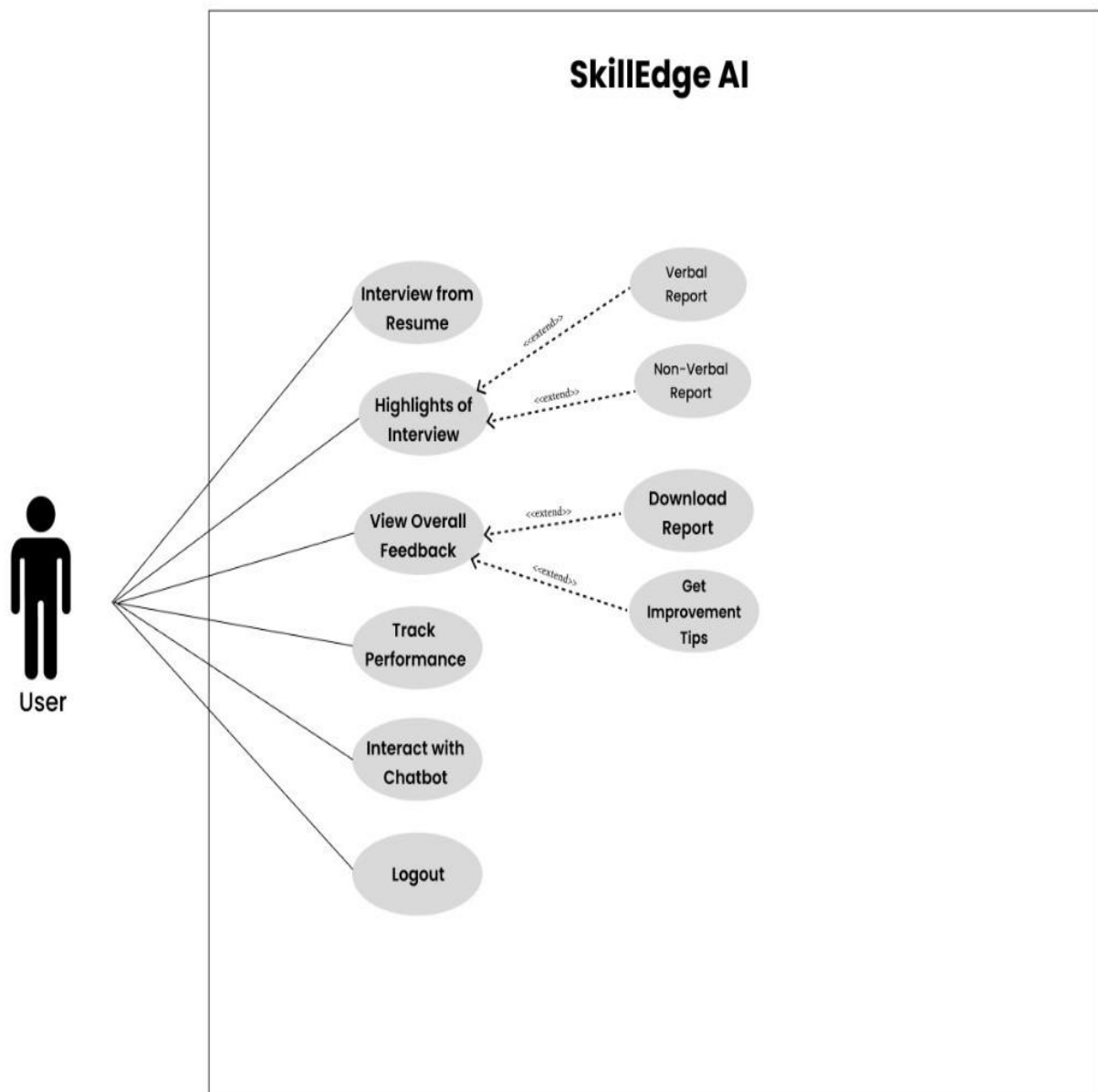


Figure 3: Use Case Diagram (2)

3.2 Use Case Description

UC -1 Signup

Table 2: Signup

Use Case ID:	UC-1
Use Case Name:	Signup
Actors:	Students, Jobseekers, Professionals
Description:	A user signs up for SkillEdge AI to access the application.
Trigger:	User indicates the desire to create a new account.
Preconditions:	None
Postconditions:	POS-1. User account is created with basic information.
Normal Flow:	1. User will select a “Sign Up” link. 2. User will enter the username, email and password. 3. The system will check whether a provided email has already been registered in the system’s database or not. 4. User will press “Sign Up” button. 5. The system will create a new account with the information provided in the system’s database.
Alternative Flows:	None
Exceptions:	In case of passwords not matching, the confirm password field will turn red with a message, "Passwords do not match." If the email is already present, a message will appear, "Email already in use; please login’’
Business Rules	None
Assumptions:	1.User has active internet Connection.

UC-2 Login

Table 3: Login

Use Case ID:	UC-2
Use Case Name:	Login
Actors:	Students, Jobseekers, Professionals
Description:	A User wants to log in to their SkillEdge AI account using their Gmail for convenience.
Trigger:	User indicates the desire to log into their account through Gmail.
Preconditions:	PRE-1. User has a registered account.
Postconditions:	POS-1. User is successfully logged in.
Normal Flow:	1. User selects "Login." 2. User enters email and password. 3. The system verifies credentials. 4. User is logged in and redirected to the dashboard.

Alternative Flows:	None.
Exceptions:	If credentials are incorrect, a message appears: "Invalid email or password."
Business Rules	BR-1: Account must be activated to log in.
Assumptions:	1. User have signed up.

UC-3 Complete Profile

Table 4: Complete Profile

Use Case ID:	UC-3
Use Case Name:	Complete Profile
Actors:	Primary Actor: User Secondary Actors: System
Description:	User provides additional details to complete their profile.
Trigger:	After signing up, the user is prompted to complete their profile.
Preconditions:	PRE-1. User must have successfully signed up.
Postconditions:	POST-1. User profile is completed and stored in the system.
Normal Flow:	1. User enters name, educational background, skills, etc. 2. User submit details. User saves the information.
Alternative Flows:	None
Exceptions:	If mandatory fields are missing, an error message appears: "Please complete all required fields."
Business Rules	None
Assumptions:	None

UC-4 Forgot Password

Table 5: Forgot Password

Use Case ID:	UC-4
Use Case Name:	Forgot Password
Actors:	Students, Jobseekers, Professionals
Description:	A User initiates the process to reset their forgotten password.
Trigger:	User indicates the desire to reset their password.
Preconditions:	PRE-1-1. User has a registered account.
Postconditions:	POS-1. User Password is resets
Normal Flow:	1. User clicks on the "Forgot Password" link. 2. The system prompts the user to enter their registered email address. 3. User enters the email address and clicks on "Submit."

	4. The system verifies the email address. 5. If the email address is valid, the system generates a unique reset link and sends it to the user's email. 6. User receives the reset link and clicks on it. 7. System validates the reset link and prompts the User to enter a new password. 8. User enters a new password and clicks on "Reset Password." The system updates the password.
Alternative Flows:	None
Exceptions:	If users Email is Invalid System displays an "Incorrect Email" error message.
Business Rules	BR-1: The email address entered must be associated with a registered account. BR-2: The reset link expires after a certain period for security reasons.
Assumptions:	None

UC-5 Update Profile

Table 6: Update Profile

Use Case ID:	UC-5
Use Case Name:	Update Profile
Actors:	Students, Jobseekers, Professionals
Description:	User manages their profile information on SkillEdge AI s.
Trigger:	User indicates the desire to log into their account.
Preconditions:	PRE-1. User is logged into their account.
Postconditions:	POS-1. Profile information is updated
Normal Flow:	1. User navigates to the profile management section. 2. User selects the information to update (e.g., Username, status, Profile picture). 3. User modifies the selected information. 4. User clicks on "Save Changes."
Alternative Flows:	None
Exceptions:	If credentials are invalid, the system displays an error message
Business Rules	BR-1: Certain profile fields cannot be empty.
Assumptions:	None

UC-6 View Profile

Table 7: View Profile

Use Case ID:	UC-6
Use Case Name:	View Profile
Actors:	Students, Jobseekers, Professionals
Description:	Users view their profile information within the SkillEdge AI platform.
Trigger:	User navigates to the profile section of the SkillEdge AI application.
Preconditions:	PRE-1. Users are logged into their SkillEdge AI account.
Postconditions:	POS-1. Profile information is displayed on the Users screen.
Normal Flow:	1. User selects the "View Profile" option from the settings menu. 2. The system retrieves and displays Users profile information, including personal details.
Alternative Flows:	None
Exceptions:	None
Business Rules	BR-1: Profile information is accessible only to the respective user.
Assumptions:	The system securely retrieves and displays accurate profile information for the User.

UC-7 Delete Profile

Table 8: UC-7: Delete Profile

Use Case ID:	UC-7
Use Case Name:	Delete Profile
Actors:	Students, Jobseekers, Professionals
Description:	Users permanently delete their profile and associated data from the SkillEdge AI platform.
Trigger:	User initiates the profile deletion process.
Preconditions:	PRE-1. User is logged into their SkillEdge AI account.
Postconditions:	POS-1. Users' profile and associated data are permanently removed.
Normal Flow:	1. User selects the "Delete Profile" option from the application menu. 2. User confirms the deletion. 3. The system permanently removes the profile and associated data from the database.
Alternative Flows:	None
Exceptions:	None
Business Rules	BR-1: Profile deletion is an irreversible process.
Assumptions:	The system securely and permanently removes the user's profile and data upon their request.

UC-8 Logout

Table 9: Logout

Use Case ID:	UC-8
Use Case Name:	Logout
Actors:	User, System
Description:	The user logs out of the SkillEdge AI system, ending the current session.
Trigger:	User initiates the logout process.
Preconditions:	PRE-1. User is logged into their SkillEdge AI account.
Postconditions:	POS-1. User is logged out and redirected to the login page.
Normal Flow:	1. User clicks on the "Logout" button or link. 2. System verifies the user's session. 3. The system terminates the session and logs out the user. 4. System redirects the user to the login page.
Alternative Flows:	• User's session has already expired. • The system automatically logs out the user and redirects them to the login page.
Exceptions:	None
Business Rules	BR-1: Logout should clear all user session data to ensure security.
Assumptions:	1. User's account remains secure after logout.

UC-9 Simulate Behavioral Interview

Table 10: Simulate Behavioral Interview

Use Case ID:	UC-9
Use Case Name	Simulate Behavioral Interview
Actors:	Students, Jobseekers and SkillEdge AI (System)
Description:	SkillEdge AI conducts a behavioral interview with a candidate, providing simulated interview experience.
Trigger:	User initiates a behavioral interview session with the SkillEdge AI.
Precondition:	PRE-1. User is logged into their account.
Postconditions:	POS-1. The simulated interview session is completed, and the SkillEdge AI provides feedback.
Normal Flow:	1. User selects to start a behavioral interview session. 2. User engages in the simulated interview, responding to behavioral questions posed by the system. 3. User give answers to questions.
Alternative Flows:	SkillEdge AI detects technical issues during the simulation.
Exceptions:	1. User Exhibits No Response 2. SkillEdge AI notifies the user to response.

	User attempts to navigate away from the test page before completing it.
Business Rules	BR-1: Simulated behavioral interview sessions adhere to a predefined time limit. BR-2: Candidates should have a stable internet connection for a seamless interview experience.
Assumptions:	The SkillEdge AI simulates realistic interview scenarios, providing valuable feedback for user improvement.

UC-10 Simulate Technical Interview

Table 11: Simulate Technical Interview

Use Case ID:	UC-10
Use Case Name:	Simulate Technical Interview
Actors:	Students, Jobseekers and SkillEdge AI (System)
Description:	SkillEdge AI conducts a technical interview with a user, utilizing audio responses for answering questions.
Trigger:	User initiates a technical interview session with the SkillEdge AI system.
Precondition:	PRE-1. User is logged into their account.
Postconditions:	POS-1. Technical interview session is completed, and detailed feedback is displayed on the user's screen.
Normal Flow:	1. User selects to start a technical interview session with default customization settings. 2. SkillEdge AI notifies the user and provides technical interview Questions. 3. User responds to technical questions verbally, using microphone button.
Alternative Flows:	1. SkillEdge AI detects unclear audio responses. • Users are prompted to clarify their response. • User Does not know the answer to the question. 2. Users select skip question. • SkillEdge AI will prompt a new question.
Exceptions:	User experiences sudden disconnection or interruption
Business Rules	BR-1: Audio responses are assessed for correctness based on predefined criteria.
Assumptions:	The SkillEdge AI effectively processes and analyzes audio input for accurate evaluation, providing a seamless technical interview experience.

UC-11 Interview Customization**Table 12: Interview Customization**

Use Case ID:	UC-11
Use Case Name:	Interview Customization
Actors:	Students, Jobseekers and SkillEdge AI (System)
Description:	Users customize the parameters of a interview simulation according to their preferences and requirements.
Trigger:	User initiates the customization process for an interview simulation.
Precondition:	PRE-1. User is logged into their account.
Postconditions:	POS-1. Technical interview simulation is customized based on user preferences.
Normal Flow:	1. User selects to customize an interview simulation. 2. SkillEdge AI prompts the user to specify preferences such as number of questions, and specific topics or skills to be covered. 3. User provides input for each customization option. 4. SkillEdge AI confirms the customization settings with the user. The system generates a customized interview simulation based on the specified preferences.
Alternative Flows:	NA
Exceptions:	User experiences sudden disconnection or interruption.
Business Rules	BR-1: Users can customize technical interview simulations within predefined limits set by the system. BR-2: Customized simulations should provide a relevant and challenging experience for the user.
Assumptions:	Users have a clear understanding of their technical interview requirements and preferences for customization.

UC-12 Upload Resume**Table 13: Upload Resume**

Use Case ID:	UC-12
Use Case Name:	Upload Resume
Actors:	Students, Jobseekers, and SkillEdge AI (System)
Description:	The SkillEdge AI system lets the user upload his resume, later on, user will be able to give an interview from his resume.
Trigger:	The user clicks the “upload resume” button and uploads his resume.
Precondition:	PRE-1: The user is logged into their account. PRE-2: The resume is in an accepted format (e.g., PDF, DOCX) and has been successfully uploaded.

Postconditions:	None
Normal Flow:	1. User logs into the SkillEdge AI system and navigates to the "Interview from Resume" module. 2. User uploads their resume using the provided upload interface. 3. SkillEdge AI stores the user resume in the data base.
Alternative Flows:	1. If the resume is in an unsupported format or contains errors, the system prompts the user to re-upload a valid resume.
Exceptions:	1. If the file upload fails due to network issues, an error message is displayed, and the upload is retried. 2. If the resume content cannot be read or processed, the system notifies the user and halts the interview session until a valid resume is provided.
Business Rules	BR-1: The resume must be processed securely, and all extracted data must be handled in compliance with data privacy standards.
Assumptions:	It is assumed that the resume extraction algorithm effectively extracts key information from standard resume formats, and that the user's resume is accurate and up to date.

UC-13 Conduct Interview from Resume

Table 14: Conduct Interview from Resume

Use Case ID:	UC-13
Use Case Name:	Conduct Interview from Resume
Actors:	Students, Jobseekers, and SkillEdge AI (System)
Description:	The SkillEdge AI system conducts an interview session tailored to the user's background by analyzing the uploaded resume. The system extracts key details (such as skills, education, and work experience) and generates relevant interview questions based on this information.
Trigger:	The user uploads their resume and initiates the "Interview from Resume" session.
Precondition:	PRE-1: The user is logged into their account. PRE-2: The resume is in an accepted format (e.g., PDF, DOCX) and has been successfully uploaded.
Postconditions:	POS-1: The tailored interview session is initiated, and the system generates domain-specific questions based on the resume. POS-2: Interview data is recorded, and the user's progress is updated accordingly.
Normal Flow:	1. User logs into the SkillEdge AI system and navigates to the "Interview from Resume" module. 2. User uploads their resume using the provided upload interface.

	3. SkillEdge AI processes the resume to extract key details (e.g., skills, education, work experience). 4. The system generates a set of interview questions tailored to the extracted information.
Alternative Flows:	1. If the resume is in an unsupported format or contains errors, the system prompts the user to re-upload a valid resume. 2. If resume processing fails (e.g., incomplete extraction), the system defaults to generating generic interview questions relevant to the user's declared domain.
Exceptions:	3. If the file upload fails due to network issues, an error message is displayed, and the upload is retried. 4. If the resume content cannot be read or processed, the system notifies the user and halts the interview session until a valid resume is provided.
Business Rules	BR-1: The resume must be processed securely, and all extracted data must be handled in compliance with data privacy standards. BR-2: Interview questions must be generated based on accurate, relevant details extracted from the resume.
Assumptions:	It is assumed that the resume extraction algorithm effectively extracts key information from standard resume formats, and that the user's resume is accurate and up to date.

UC-14 Verbal Response Evaluation

Table 15: Verbal Response Evaluation

Use Case ID:	UC-14
Use Case Name:	Verbal Response Evaluation
Actors:	Students, Jobseekers, and SkillEdge AI (System)
Description:	After completing an interview session, the SkillEdge AI system provides a detailed performance report that includes separate sections for verbal and non-verbal feedback. The verbal feedback module specifies whether the answers given by user are correct or not.
Trigger:	The user selects the "View Verbal Feedback" option after the interview session is complete and the verbal evaluated report is generated.
Precondition:	PRE-1: The interview session has been completed, and all relevant audio evaluation data is available.
Postconditions:	POS-1: The user can view and download the report.

Normal Flow:	1. The user completes the interview session, and the overall performance report is generated. 2. The user selects the "View Verbal Feedback" option. 3. The system retrieves and analyzes the recorded audio data from the interview, focusing on answers given by users. 4. The system generates and displays a detailed verbal feedback section, including numerical scores and graphical representations.
Alternative Flows:	1. If the audio data is incomplete or unclear (e.g., due to background noise), the system notifies the user and suggests adjusting microphone settings or reattempting the session.
Exceptions:	In the event of a processing error or if the audio data cannot be analyzed due to technical issues, the system displays an error message and advises the user to retry the session or check their device settings.
Business Rules	BR-1: verbal feedback is generated only when sufficient and clear audio data is captured by the system.
Assumptions:	It is assumed that the system's audio analysis module reliably captures and processes speech data using the user's microphone, and that environmental factors (such as background noise) do not significantly interfere with the accuracy of the analysis.

UC-15 Non-Verbal Response Evaluation

Table 15: Non-Verbal Evaluation

Use Case ID:	UC-15
Use Case Name:	Non-Verbal Response Evaluation
Actors:	Students, Jobseekers, and SkillEdge AI (System)
Description:	After completing an interview session, the SkillEdge AI system provides a detailed performance report that includes separate sections for verbal and non-verbal feedback. The non-verbal feedback module specifically analyzes audio cues such as speech rate, filler word usage (e.g., "um," "uh"), tone, and pitch to help users improve their vocal delivery and overall communication style.
Trigger:	The user selects the "View Non-Verbal Feedback" option after the interview session is complete and the non-verbal evaluated report is generated.
Precondition:	PRE-1: The interview session has been completed, and all relevant audio evaluation data is available. PRE-2: The user has accessed the overall performance report
Postconditions:	POS-1: The non-verbal (audio) feedback is displayed as a distinct section within the overall performance report. POS-2: The user can view and download detailed analysis on speech rate, filler words, tone, and pitch, which is stored in the system for

	future reference and progress tracking.
Normal Flow:	1. The user completes the interview session, and the overall performance report is generated. 2. The user selects the "View Non-Verbal Feedback" option. 3. The system retrieves and analyzes the recorded audio data from the interview, focusing on speech rate, filler word usage, tone, and pitch. 4. The system generates and displays a detailed non-verbal (audio) feedback section, including numerical scores and graphical representations.
Alternative Flows:	1. If the audio data is incomplete or unclear (e.g., due to background noise), the system notifies the user and suggests adjusting microphone settings or reattempting the session.
Exceptions:	In the event of a processing error or if the audio data cannot be analyzed due to technical issues, the system displays an error message and advises the user to retry the session or check their device settings.
Business Rules	1. BR-4: Non-verbal (audio) feedback is generated only when sufficient and clear audio data is captured by the system.
Assumptions:	It is assumed that the system's audio analysis module reliably captures and processes speech data using the user's microphone, and that environmental factors (such as background noise) do not significantly interfere with the accuracy of the analysis.

UC-16 View Verbal/Non-Verbal Report

Table 16: View Verbal/Non-Verbal Report

Use Case ID:	UC-16
Use Case Name:	View Verbal/Non-Verbal Report.
Actors:	Students, Jobseekers, and SkillEdge AI (System)
Description:	After the interview session is completed, user can access the verbal and non-verbal reports reflecting his performance, separately.
Trigger:	The user clicks either on "View Verbal Response" or "View Non-Verbal Response" buttons at the end of the interview session.
Precondition:	PRE-1: The interview session has been completed. PRE-2: All evaluation data (audio analysis, question responses, and non-verbal metrics) is available and stored in the system.
Postconditions:	POS-1: A comprehensive performance report is generated and displayed on the user's screen.
Normal Flow:	1. User completes the interview session. 2. User clicks the "View Verbal Response" or "View Non-Verbal Response" button.

	3. SkillEdge AI aggregates evaluation data from the interview session. 4. A detailed report is generated and displayed on-screen.
Alternative Flows:	None
Exceptions:	1. If an error occurs during data aggregation or report generation (e.g., file generation failure), the system displays an error message. 2. In case of a network or system interruption, the report generation is paused and resumed once connectivity is restored.
Business Rules	BR-1: The performance report must include both quantitative metrics (e.g., scores, time metrics) and qualitative feedback (e.g., strengths, improvement suggestions).
Assumptions:	It is assumed that the interview evaluation process has successfully captured all necessary data, and that the system has adequate processing power to compile and generate the performance report without errors.

UC-17 Access Chatbot for Report-Based Querying

Table 18: Access Chatbot for Report-Based Querying

Use Case ID:	UC-17
Use Case Name:	Access Chatbot for Report-Based Querying
Actors:	Students, Jobseekers, and SkillEdge AI (System)
Description:	After the performance report is generated, the SkillEdge AI system provides an interactive chatbot that allows users to ask questions related to their evaluation report. The chatbot accesses the report data and offers personalized guidance, suggestions, and answers tailored to help users improve their interview performance.
Trigger:	The user clicks the "Chat with Report Assistant" option after viewing their performance report.
Precondition:	PRE-1: The performance report has been generated and is available. PRE-2: The user is logged into the system.
Postconditions:	POS-1: The chatbot provides contextual, report-based responses and guidance. POS-2: The chatbot session is logged for further analysis and continuous improvement of the system.

Normal Flow:	1. The user selects the chatbot option from the performance report screen. 2. The system opens an interactive chatbot interface. 3. The user types a question related to their report (e.g., "Why was my technical response rated low?"). 4. The chatbot retrieves and analyzes the relevant sections of the performance report. 5. The chatbot responds with detailed explanations and actionable suggestions. 6. The user may ask follow-up questions for further clarification.
Alternative Flows:	1. If the chatbot does not fully understand the query, it prompts the user for additional clarification or offers general tips based on the overall report data. 2. The user may choose to end the chatbot session at any time, which will then be logged as a completed interaction.
Exceptions:	If the system encounters an error retrieving the performance report data, the chatbot displays an error message and advises the user to try again later.
Business Rules	BR-1: The chatbot must provide responses that are directly relevant to the contents of the performance report. BR-2: All chatbot responses should be concise, actionable, and maintain a supportive tone.
Assumptions:	It is assumed that the performance report data is structured and stored in a way that allows the chatbot to access and interpret it effectively, ensuring that all responses are accurate and personalized.

UC-18 Access Chatbot for Resume-Based Querying

Table 19 Access Chatbot for Report-Based Querying

Use Case ID:	UC-18
Use Case Name:	Access Chatbot for Report-Based Querying
Actors:	Students, Jobseekers, and SkillEdge AI (System)
Description:	Users interact with a chatbot to ask questions specific to their uploaded resume (e.g., projects, skills, experience). The chatbot analyzes the resume content and provides personalized suggestions for optimization, ATS compliance, or interview preparation.
Trigger:	The user clicks "Ask Resume Assistant" from the resume dashboard.
Precondition:	PRE-1: The user has uploaded and parsed a resume. PRE-2: The user is logged into the system.
Postconditions:	POS-1: The chatbot provides resume-specific responses and suggestions. POS-2: The interaction is logged for improving resume analysis.

Normal Flow:	1. The system opens the chatbot interface with access to the user's parsed resume data. 2. The user asks a resume-related question (e.g., "What skills should I highlight?" or "How can I improve my project descriptions?"). 3. The chatbot analyzes the resume's content (e.g., skills, job roles, projects). 4. The chatbot responds with tailored suggestions (e.g., "Add quantifiable results to Project X" or "Include the keyword 'Agile' for ATS compatibility").
Alternative Flows:	1. If the chatbot cannot interpret the query (e.g., ambiguous project name), it asks the user to specify (e.g., "Are you referring to your 'E-commerce Dashboard' project?"). 2. If the user asks, "Is my resume format good?", the chatbot evaluates structure and suggests improvements (e.g., "Use bullet points for readability").
Exceptions:	1. If no resume is found, the chatbot responds: "Please upload your resume first to use this feature."
Business Rules	BR-1: The chatbot must provide responses that are directly relevant to the contents of the performance report. BR-2: All chatbot responses should be concise, actionable, and maintain a supportive tone.
Assumptions:	It is assumed that the performance report data is structured and stored in a way that allows the chatbot to access and interpret it effectively, ensuring that all responses are accurate and personalized.

UC-19 View Overall Feedback

Table 20: View Overall Feedback

Use Case ID:	UC-19
Use Case Name:	View overall Feedback
Actors:	System, User
Description:	Users receive feedback on their performance during an interview session within SkillEdge AI.
Trigger:	User initiates the interview
Precondition:	PRE-1. User is logged into SkillEdge AI and has initiated an interview.
Postconditions:	POS-1. Feedback is provided right after the interview session.

Normal Flow:	1. User initiates an interview. 2. The system continuously analyzes the user's responses and answers. 3. The system provides feedback.
Alternative Flows:	None
Exceptions:	During the session, feedback may not be accurate in cases of technical issues.
Business Rules	BR-1: feedback is provided based on real-time analysis of user interactions and performance.
Assumptions:	The interview session is conducted in a suitable environment.

UC-20 Download Evaluation Report

Table 21: View Progress Report

Use Case ID:	UC-20
Use Case Name:	Download Evaluation Report
Actors:	Students, Jobseekers, and SkillEdge AI (System)
Description:	The SkillEdge AI system allows users to download their detailed evaluation report after completing a technical interview session.
Trigger:	The user clicks the "Download Report" button after the technical interview session is complete.
Precondition:	PRE-1: The technical interview session has been completed, and an evaluation report has been generated.
Postconditions:	POS-1: The evaluation report is successfully downloaded to the user's device and the system updates the user's progress history with the download timestamp.
Normal Flow:	1. The user completes the technical interview session. 2. SkillEdge AI generates a detailed evaluation report. 3. The system displays a "Download Report" option to the user. 4. The user clicks the "Download Report" button.
Alternative Flows:	If SkillEdge AI detects an issue during report generation, the system displays an error message and prompts the user to try again.
Exceptions:	None
Business Rules	BR-1: The evaluation report must include detailed feedback, scores, and graphical representations of performance metrics.
Assumptions:	It is assumed that the SkillEdge AI system can correctly compile evaluation data into a standard downloadable format and that the user's device supports file downloads without issues.

UC-21 View Progress Report**Table 22: View Progress Report**

Use Case ID:	UC-21
Use Case Name:	View Progress Reports
Actors:	User, SkillEdge AI
Description:	Users review their interview performance progress reports within the SkillEdge AI platform.
Trigger:	User selects the "View Progress Reports" option from the application menu.
Precondition:	PRE-1. User is logged into their SkillEdge AI account. PRE-2. User has stimulated an interview.
Postconditions:	POS-1. Progress reports detailing interview performance are displayed on the user's screen.
Normal Flow:	1. User selects the "View Progress Reports" option from the application menu. 2. User Select "View Behavioral Interview Report" option from the menu. 3. System retrieves and displays progress reports, including performance metrics such as speech and grammar analysis Speech quality and confidence, strengths, and areas for improvement.
Alternative Flows:	None
Exceptions:	Error, you have not stimulated any interview.
Business Rules	BR-1: Progress reports are accessible only to the respective candidate.
Assumptions:	The system accurately compiles and presents progress information based on interview performance data.

3.3 Event Response Table

Table 23: Partial Event-Response Table for SkillEdge AI

Event	System State	Response	Exceptions
User logs in	Authenticating user	The system verifies the credentials and grants access to the dashboard.	Invalid username/password or account not found.
User selects interview type and initiates session	Waiting for interview session initiation	The system displays interview instructions.	Invalid selection, session initialization errors, or network issues.
System generates interview questions	Generating dynamic questions	The system generates domain-specific interview questions.	Question generation failure or model error.
User provides verbal response	Recording audio response	The system captures the user's audio response and confirms the recording.	Microphone issues, unclear audio, or high ambient noise.
System evaluates verbal response	Evaluating verbal content	The system processes the audio response using NLP techniques to assess answer correctness, clarity, and overall coherence.	Processing errors or poor audio quality.
System evaluates non-verbal aspects of audio response	Analyzing vocal delivery	The system analyzes speech rate, tone, pitch, and filler word frequency to provide non-verbal feedback on the user's delivery.	Poor audio quality or inconsistent input.
System compiles performance report	Aggregating evaluation data	The system aggregates metrics from both verbal and non-verbal evaluations to generate a comprehensive performance report.	Incomplete data or aggregation errors.
User views performance report	Displaying evaluation report	The system presents a detailed report including both verbal and non-verbal feedback, complete with scores and graphical visualizations.	Report display errors or missing data.

User downloads performance report	Report download initiated	The system converts the report to PDF/DOCX format and downloads it to the user's device.	File generation failure or download errors.
User interact with context aware chatbot	Chatbot session initiated	The system opens an interactive chatbot interface, allowing the user to ask questions about their performance report and receive contextual feedback.	Chatbot connectivity issues or query processing errors.
User logs out	Ending session	Session is terminated.	Session termination error or network timeout.

4. Functional Requirements

4.1 UC-1 Sign Up:

The following are the functional requirements for the Signup use case.

4.1.1 Signup.DisplayForm

Identifier	FR-4.1.1
Title	Signup.DisplayForm
Requirement	The user shall be able to create an account by providing necessary information through the signup process, including username, email, and password.
Source	Raees
Rationale	To enable users to access the SkillEdge AI platform and utilize its features, a user account is required.
Business Rule	BR-1 One Email Address can only have 1 SkillEdge AI user Account
Dependencies	None
Priority	High

4.1.2 Signup.EnterName

Identifier	FR-4.1.2
Title	Signup.EnterName
Requirement	The user shall be able to Enter Email Address to Create a user account.
Source	Raees
Rationale	To enable users to access the SkillEdge AI platform and utilize its features, a user account is required.
Business Rule	BR-1 One Email Address can only have 1 SkillEdge AI user Account
Dependencies	FR-4.1.1
Priority	High

4.1.3 Signup.VerifyUniqueUsername

Identifier	FR-4.1.3
Title	Signup.VerifyUniqueUsername
Requirement	The system shall verify that the username chosen by a user during registration or modification is unique and not already in use by another user.
Source	Student/Jobseeker
Rationale	To ensure that each user has a distinct and unique username for identification purposes.
Business Rule	BR-1: Each user must have a unique username.
Dependencies	FR-4.1.2
Priority	High

4.1.4 Signup.EnterEmailAddress

Identifier	FR-4.1.4
Title	Signup.EnterEmailAddress
Requirement	The user shall be able to Enter Email Address to Create a user account.
Source	Raees
Rationale	To enable users to access the SkillEdge AI platform and utilize its features, a user account is required.
Business Rule	BR-1 One Email Address can only have 1 SkillEdge AI user Account
Dependencies	FR-4.1.1
Priority	High

4.1.5 Signup.EmailUnique

Identifier	FR-4.1.5
Title	Signup.EmailUnique
Requirement	The system shall validate that the entered email address is unique during registration.
Source	Raees Zakir
Rationale	Prevent duplicate accounts with the same email.
Business Rule	BR-1 Strong passwords to ensure privacy of user data.
Dependencies	FR-4.1.4
Priority	High

4.1.6 Signup.EnterPassword

Identifier	FR-4.1.6
Title	Signup.EnterPassword
Requirement	The user shall be able to create an account by providing a Strong password with Uppercase and lowercase Characters.
Source	Raees
Rationale	To enable users to access the SkillEdge AI platform and utilize its features, a user account is required.
Business Rule	BR-1 Strong passwords to ensure privacy of user data.
Dependencies	FR-4.1.1
Priority	High

4.1.7 Signup.CheckPassword

Identifier	FR-4.1.7
Title	Signup.CheckPassword
Requirement	The system shall validate that the entered password meets security requirements (e.g., length, special characters).
Source	Raees Zakir
Rationale	Ensure password strength for user accounts.
Business Rule	BR-1 Strong passwords to ensure privacy of user data.
Dependencies	FR-4.1.6
Priority	High

4.1.8 Signup.PasswordMismatch

Identifier	FR-4.1.8
Title	Signup.PasswordMismatch
Requirement	If the password and confirm password do not match, the system shall display an error message and prompt the user to re-enter both fields.
Source	Raees Zakir
Rationale	Ensure user provides the correct password.
Business Rule	BR-1 Strong passwords to ensure privacy of user data.
Dependencies	FR-4.1.6, FR-4.1.7
Priority	High

4.1.9 Signup.UsernameUnique

Identifier	FR-4.1.9
Title	Signup.UsernameUnique
Requirement	The system shall validate that the entered username is unique.
Source	Raees Zakir
Rationale	Ensure unique usernames for accounts.
Business Rule	BR-1 Strong passwords to ensure privacy of user data.
Dependencies	FR-4.1.2
Priority	High

4.1.10 Signup.UsernameExists

Identifier	FR-4.1.12
Title	Signup.UsernameExists
Requirement	If the entered username is already taken, the system shall prompt the user to choose a different username.
Source	Raees Zakir
Rationale	Prevent duplicate usernames and improve user experience.
Business Rule	BR-1
Dependencies	FR-4.1.9
Priority	High

4.2 UC-2 Login:

The following are the functional requirements for the Login use case.

4.2.1 Login.DisplayForm

Identifier	FR-4.2.1
Title	Login.DisplayForm
Requirement	When the user selects the "Login" option, the system shall display a login form with fields for email and password.
Source	Farrukh
Rationale	Users need to provide credentials to access their accounts.
Business Rule	None
Dependencies	None
Priority	High

4.2.2 Login.EnterCredentials

Identifier	FR-4.2.2
Title	Login.EnterCredentials
Requirement	The user shall be able to enter their email and password in the login form.
Source	Farrukh
Rationale	Users need to enter their credentials for authentication.
Field details	The form shall include fields: Email and Password.
Business Rule	None
Dependencies	FR-4.2.1
Priority	High

4.2.3 Login.ValidateEmailFormat

Identifier	FR-4.2.3
Title	Login.ValidateEmailFormat
Requirement	The system shall validate that the entered email follows a correct email format (e.g., example@domain.com).
Source	Farrukh
Rationale	Prevent invalid email formats during login attempts.
Business Rule	None
Dependencies	FR-4.2.2
Priority	High

4.2.4 Login.VerifyCredentials

Identifier	FR-4.2.4
Title	Login.VerifyCredentials
Requirement	The system shall verify that the provided email and password match an existing user account.
Source	Farrukh
Rationale	Ensure only registered users can access the system.
Business Rule	None
Dependencies	FR-4.2.3
Priority	High

4.2.5 Login.InvalidCredentials

Identifier	FR-4.2.5
Title	Login.InvalidCredentials
Requirement	If the entered email or password is incorrect, the system shall display an error message: "Invalid email or password."
Source	Farrukh
Rationale	Provide feedback to users for incorrect login attempts.
Business Rule	None
Dependencies	FR-4.2.4
Priority	High

4.2.6 Login.SuccessfulLogin

Identifier	FR-4.2.6
Title	Login.SuccessfulLogin
Requirement	If credentials are valid, the system shall log the user in and redirect them to the dashboard.
Source	Farrukh
Rationale	Allow authorized users to access their profiles.
Business Rule	None
Dependencies	FR-4.2.4
Priority	High

4.2.7 Login.SessionManagement

Identifier	FR-4.2.7
Title	Login.SessionManagement
Requirement	The system shall maintain a session for the logged-in user until they log out or the session expires.
Source	Farrukh
Rationale	Ensure continuous user access without frequent logins.
Business Rule	None
Dependencies	FR-4.2.6
Priority	High

4.2.8 Login.RememberMeOption

Identifier	FR-4.2.8
Title	Login.RememberMeOption
Requirement	If the user selects "Remember Me," the system shall keep the user logged in for future sessions unless they manually log out.
Source	Farrukh
Rationale	Improve user experience by avoiding repeated logins.
Business Rule	None
Dependencies	FR-4.2.6
Priority	Medium

4.2.9 Login.LockAccount

Identifier	FR-4.2.10
Title	Login.LockAccount
Requirement	The system shall temporarily lock the account after 5 consecutive failed login attempts and notify the user.
Source	Farrukh
Rationale	Prevent brute-force attacks on user accounts.
Business Rule	None
Dependencies	FR-4.2.5
Priority	High

4.3 UC-3 Complete Profile

The following are the functional requirements for the Complete Profile use case.

4.3.1 CompleteProfile.EnterDetails

Identifier	FR-4.3.1
Title	CompleteProfile.Enter Details
Requirement	The system shall provide fields for the user to enter their details like technical background, education etc.
Source	Raees Zakir
Rationale	Users need to enter their details to complete their profile.
Business Rule	None
Dependencies	None
Priority	High

4.3.2 CompleteProfile.SubmitButton

Identifier	FR-4.3.2
Title	CompleteProfile.SubmitButton
Requirement	The system shall provide a "Submit" button for users to submit their profile details.
Source	Raees Zakir
Rationale	Users need a way to confirm and submit their entered details.
Business Rule	None
Dependencies	FR-4.3.1
Priority	High

4.3.3 CompleteProfile.Validation

Identifier	FR-4.3.3
Title	CompleteProfile.Validation
Requirement	The system shall validate that all mandatory fields are filled before allowing submission.
Source	Raees Zakir
Rationale	Ensures the user provides all necessary details before proceeding.
Business Rule	None
Dependencies	FR-4.3.1, FR-4.3.2
Priority	High

4.3.4 CompleteProfile.ErrorHandling

Identifier	FR-4.3.4
Title	CompleteProfile.ErrorHandling
Requirement	If mandatory fields are missing, the system shall display an error message: "Please complete all required fields."
Source	Raees Zakir
Rationale	Helps users identify and correct missing information.
Business Rule	None
Dependencies	FR-4.3.3
Priority	High

4.3.5 CompleteProfile.SaveInformation

Identifier	FR-4.3.5
Title	CompleteProfile.SaveInformation
Requirement	The system shall store the completed profile information in the database after successful submission.
Source	Raees Zakir
Rationale	Ensures the user's profile is saved for future use.
Business Rule	None
Dependencies	FR-4.3.2, FR-4.3.3
Priority	High

4.3.6 CompleteProfile.SuccessMessage

Identifier	FR-4.3.5
Title	CompleteProfile.SuccessMessage
Requirement	Upon successful profile submission, the system shall display a success message: "Your profile has been successfully completed."
Source	Raees Zakir
Rationale	Provides confirmation that the profile has been saved successfully.
Business Rule	None
Dependencies	FR-4.3.2, FR-4.3.3
Priority	High

4.4 UC-4 Update Profile

The following are the functional requirements for Update Profile Use Case

4.4.1 UpdateProfile.Access

Identifier	FR-4.4.1
Title	UpdateProfile.Access
Requirement	The system shall allow users to access the "Edit Profile" section to modify their profile information.
Source	Raees Zakir
Rationale	Users should be able to update their personal information when needed.
Business Rule	None
Dependencies	None
Priority	High

4.4.2 UpdateProfile.ModifyDetails

Identifier	FR-4.4.2
Title	UpdateProfile.ModifyDetails
Requirement	The system shall allow users to modify their profile details such as name, email, skills etc.
Source	Raees Zakir
Rationale	Users may need to update their information due to changes in personal details.
Business Rule	None
Dependencies	FR-4.4.1
Priority	High

4.4.3 UpdateProfile.SubmitChanges

Identifier	FR-4.4.3
Title	UpdateProfile.SubmitChanges
Requirement	The system shall provide a "Save" button to allow users to submit profile changes.
Source	Raees Zakir
Rationale	Users need a way to confirm their modifications.
Business Rule	None
Dependencies	FR-4.4.2
Priority	High

4.4.4 UpdateProfile.Validation

Identifier	FR-4.4.4
Title	UpdateProfile.Validation
Requirement	The system shall validate that all required fields are filled before updating the profile.
Source	Raees Zakir
Rationale	Ensures the integrity and completeness of user data.
Business Rule	None
Dependencies	FR-4.4.3
Priority	High

4.4.5 UpdateProfile.ConfirmationMessage

Identifier	FR-4.4.5
Title	UpdateProfile.ConfirmationMessage
Requirement	The system shall display a confirmation message after a successful profile update.
Source	Raees Zakir
Rationale	Provides feedback to the user that the update was successful.
Business Rule	None
Dependencies	FR-4.4.4
Priority	Medium

4.4.6 UpdateProfile.DataConsistency

Identifier	FR-4.4.6
Title	UpdateProfile.DataConsistency
Requirement	The system shall ensure that updated profile information is reflected accurately across the system.
Source	Raees Zakir
Rationale	Prevents inconsistencies in user data.
Business Rule	None
Dependencies	FR-4.4.5
Priority	High

4.5 UC-5 View Profile

The following are the functional requirements for View Profile Use Case

4.5.1 ViewProfile.Access

Identifier	FR-4.5.1
Title	ViewProfile.Access
Requirement	The system shall allow the user to access their profile by clicking on "Profile" in the menu.
Source	Farrukh
Rationale	Users need to access their profile to view personal information.
Business Rule	None
Dependencies	FR-4.4.1
Priority	High

4.5.2 View Profile.DisplayInformation

Identifier	FR-4.5.2
Title	ViewProfile.Display Information
Requirement	The system shall retrieve and display the user's profile information, including name, email, and other details.
Source	Farrukh
Rationale	Users should be able to view their stored profile data.
Business Rule	None
Dependencies	FR-4.4.1
Priority	High

4.5.3 View Profile.DataConsistency

Identifier	FR-4.5.3
Title	ViewProfile.DataConsistency
Requirement	The system shall ensure that the profile data displayed is up-to-date and consistent with stored information.
Source	Farrukh
Rationale	Users should see accurate and current profile information.
Business Rule	None
Dependencies	FR-4.4.2
Priority	High

4.6 UC-6 Delete Account

The following are the functional requirements for Delete Account Use Case

4.6.1 DeleteAccount.Confirmation

Identifier	FR-4.6.1
Title	DeleteAccount.Confirmation
Requirement	The system shall prompt the user for confirmation before deleting the account.
Source	Raees Zakir
Rationale	Prevents accidental account deletion.
Business Rule	None
Dependencies	None
Priority	High

4.6.2 DeleteAccount.DataRemoval

Identifier	FR-4.6.2
Title	DeleteAccount.DataRemoval
Requirement	The system shall permanently remove all user data upon account deletion.
Source	Raees Zakir
Rationale	Ensures compliance with privacy policies.
Business Rule	None
Dependencies	FR-4.6.1
Priority	High

4.6.3 DeleteAccount.logout

Identifier	FR-4.6.3
Title	DeleteAccount.logout
Requirement	The system shall automatically log out the user after account deletion.
Source	Raees Zakir
Rationale	Ensures security by ending user sessions.
Business Rule	None
Dependencies	FR-4.6.2
Priority	High

4.7 UC-7 Forget Password

The following are the functional requirements for Forget Password Use Case

4.7.1 ForgetPassword.RequestPasswordReset

Identifier	FR-4.7.1
Title	ForgetPassword.Request Password Reset
Requirement	The system shall provide users with an option to request a password reset by entering their registered email or phone number.
Source	Raees
Rationale	To let users regain access to their accounts in case of forgotten password.
Business Rule	BR-1 Ensure User Privacy and Integrity.
Dependencies	None
Priority	High

4.7.2 ForgetPassword.VerifyIdentityviaOTP

Identifier	FR-4.7.2
Title	ForgetPassword.VerifyIdentityviaOTP
Requirement	The system shall send a one-time password (OTP) to the user's registered email or phone number for identity verification.
Source	Raees
Rationale	To confirm user identity before allowing a password reset.
Business Rule	BR-1 Ensure User Privacy and integrity.
Dependencies	FR-4.7.1 (Request Password Reset)
Priority	High

4.7.3 ForgetPassword.ResetPassword

Identifier	FR-4.7.2
Title	ForgetPassword.ResetPassword
Requirement	The system shall allow users to enter a new password after successfully verifying their identity via OTP.
Source	Raees
Rationale	To ensure secure account recovery by allowing users to set a new password.
Business Rule	BR-1 Ensure User Privacy and integrity.
Dependencies	FR-4.7.1 (Request Password Reset)
Priority	High

4.8 UC-8 Logout

The following are the functional requirements for Logout Use Case

4.8.1 Logout.Initiate

Identifier	FR-4.8.1
Title	Logout.Initiate
Requirement	The system shall provide a "Logout" button for users to initiate the logout process.
Source	Farrukh
Rationale	Users need a clear way to log out of the system.
Business Rule	None
Dependencies	None
Priority	High

4.8.2 Logout.Session Termination

Identifier	FR-4.8.2
Title	Logout.SessionTermination
Requirement	The system shall terminate the user session upon logout.
Source	Farrukh
Rationale	Ensures security by preventing unauthorized access to an active session.
Business Rule	None
Dependencies	FR-4.8.1
Priority	High

4.8.3 Logout.Redirect

Identifier	FR-4.8.3
Title	Logout.Redirect
Requirement	The system redirects the user to the login page after logout.
Source	Farrukh
Rationale	Provides a clear transition after logout.
Business Rule	None
Dependencies	FR-4.8.2
Priority	Medium

4.9 UC-9 Simulate Behavioral Interview

The following are the functional requirements for Simulate Behavioral Interview Use Case

4.9.1 SimulateBehavioralInterview.UserInterface

Identifier	FR-4.9.1
Title	SimulateBehavioralInterview.UserInterface
Requirement	The system shall allow a registered user to simulate a behavioral Interview.
Source	Raees
Rationale	To provide users with a dedicated feature for conducting behavioral interviews
Business Rule	The platform must support the conduct of behavioral interviews
Dependencies	FR-4.2.2 (Login)
Priority	High

4.9.2 SimulateBehavioralInterview.InsightsIntoUserTraits

Identifier	FR-4.9.2
Title	SimulateBehavioralInterview.InsightsIntoUserTraits
Requirement	The system shall analyze user traits and behavior, extracting insights to supplement human evaluations during interviews.
Source	Farrukh
Rationale	Enhance the evaluation process by combining human judgment with data-driven insights.
Business Rule	User data analysis adheres to privacy regulations.
Dependencies	FR-4.9.1 (Conduct Behavioral Interview)
Priority	High

4.9.3 SimulateBehavioralInterview.AudioUnclearPrompt

Identifier	FR-4.9.3
Title	SimulateBehavioralInterview.AudioUnclearPrompt
Requirement	System shall prompt user to speak more clearly, if the user's speech is unclear.
Source	Raees
Rationale	Ensure accurate speech recognition and improve user interaction.
Business Rule	User data analysis adheres to privacy regulations.
Dependencies	FR-4.9.1(Conduct Behavioral Interview)
Priority	High

4.10 UC-10 Simulate Technical Interview

The following are the functional requirements for Simulate Technical Interview Use Case

4.10.1 Simulate Technical Interview.Interface

Identifier	FR-4.10.1
Title	SimulateTechnicalInterview.ConductTechnicalInterview
Requirement	A registered User shall be able to conduct technical interviews on the Application.
Source	Farrukh
Rationale	To provide users with a dedicated feature for conducting technical interviews, enabling a focused evaluation of technical skills.
Business Rule	The platform must support the conduct of technical interviews with a focus on technical skills.
Dependencies	FR-4.2.2 (Login)
Priority	High

4.10.2 Simulate Technical Interview.Set Topic for the Interview

Identifier	FR-4.10.2
Title	SimulateTechnicalInterview.InterviewTopicsPrompt
Requirement	The system shall prompt the user to select the interview topics of their choice
Source	Raees
Rationale	To enhance the interview experience by offering users multiple choices.
Business Rule	None
Dependencies	FR-4.10.1 (Technical Interview)
Priority	Medium

4.10.3 Simulate Technical Interview.Interview Questions

Identifier	FR-4.10.3
Title	SimulateTechnicalInterview.InterviewQuestions
Requirement	The system shall provide customized simulations tailored to the user's skills and knowledge level for a relevant and challenging experience.
Source	Raees
Rationale	Enhance engagement and learning through personalized simulation.
Business Rule	None
Dependencies	FR-4.10.1 (Technical Interview)
Priority	Medium

4.11 UC-11 Interview Customization

The following are the functional requirements for Interview Customization Use Case

4.11.1 InterviewCustomization.InterviewDurationTime

Identifier	FR-4.11.1
Title	InterviewCustomization.InterviewDurationTime
Requirement	The user shall be able to set the duration of the interview.
Source	Farrukh
Rationale	To enhance the interview experience by offering users multiple choices.
Business Rule	None
Dependencies	None
Priority	Medium

4.11.2 InterviewCustomization.SetNumbersofQuestions

Identifier	FR-4.11.2
Title	InterviewCustomization.SetNumberOfQuestions
Requirement	The system shall prompt the user to enter number of questions to be asked.
Source	Farrukh
Rationale	To enhance the interview experience by offering users multiple choices.
Business Rule	None
Dependencies	None
Priority	Medium

4.11.3 InterviewCustomization.ShowRemainingTime

Identifier	FR-4.11.3
Title	InterviewCustomization.ShowRemainingTime
Requirement	The system shall prompt the user during the Interview about Time left to complete his answer.
Source	Farrukh
Rationale	To enhance the interview experience.
Business Rule	BR-1 Total time required to answer a question is 2 minutes.
Dependencies	FR-4.10.1
Priority	Medium

4.12 UC-12: Upload Resume

The following are the functional requirements for Interview Customization Use Cas

4.12.1 UploadResume.ResumeUploading & Parsing

Identifier	FR-4.12.1
Title	UploadResume.Resume Upload and Parsing
Requirement	The system shall allow users to upload their resumes (supported formats: PDF, DOCX) and automatically extract key details such as skills and education.
Source	Raees
Rationale	To tailor interview questions based on the user's professional background.
Business Rule	The system must support specified file formats.
Dependencies	None
Priority	High

4.12.2 UploadResume.InformationExtraction

Identifier	FR-4.12.2
Title	UploadResume.InformationExtraction
Requirement	The system shall be able to extract information like profession, skills, interests and store the information.
Source	Raees
Rationale	To provide a more personalized and relevant interview experience that reflects the user's background.
Business Rule	Generated questions must comprehensively cover resume details.
Dependencies	FR-4.12.1
Priority	High

4.12.3 Upload Resume.Edit Information

Identifier	FR-4.12.3
Title	UploadResume.EditInformation
Requirement	The system allows users to edit resume information directly on the platform.
Source	Raees
Rationale	To provide interview questions based on the user's choice.
Business Rule	The system must support specified file formats.
Dependencies	FR-4.12.1
Priority	Medium

4.13 UC-13: Interview From Resume

The following are the functional requirements for interview from resume Use Case

4.13.1 InterviewfromResume.QuestionfromSkillSet

Identifier	FR-4.13.1
Title	InterviewfromResume.QuestionFromSkillSet
Requirement	The system shall generate interview questions based on information extracted from the uploaded resume, specifically covering the user's skill set.
Source	Raees
Rationale	To provide a personalized and relevant interview experience.
Business Rule	The generated questions must cover the skills listed in the resume.
Dependencies	FR-4.12.2
Priority	High

4.13.2 InterviewfromResume.QuestionfromProjects

Identifier	FR-4.13.2
Title	InterviewfromResume.QuestionFromProjects
Requirement	The system shall generate interview questions based on project experience details extracted from the uploaded resume.
Source	Raees
Rationale	To assess the user's project-based experience, problem-solving skills.
Business Rule	The generated questions must comprehensively cover projects mentioned in the resume.
Dependencies	FR-4.12.2
Priority	High

4.13.3 InterviewfromResume.ConductInterview

Identifier	FR-4.13.3
Title	InterviewfromResume.ConductInterview
Requirement	The system shall conduct an interview based on questions generated from the uploaded resume, ensuring a structured and relevant interview process.
Source	Raees
Rationale	To simulate real interview scenarios based on the user's resume content, ensuring practical and targeted interview practice.
Business Rule	The interview must dynamically adjust based on the user's responses and maintain a structured flow covering all key resume aspects.
Dependencies	FR-4.13.1, FR-4.13.2
Priority	High

4.14 UC-14: Evaluate Verbal Response

The following are the functional requirements for Evaluate Verbal Response Use Case

4.14.1 EvaluateVerbalResponse.EvaluateAnswer

Identifier	FR-4.14.1
Title	EvaluateVerbalResponse.EvaluateAnswer
Requirement	The system shall analyze the users answers, and evaluate whether they are right or wrong
Source	Farrukh
Rationale	To ensure that users give right answers during the interview.
Business Rule	The platform must evaluate answers with high accuracy.
Dependencies	FR-4.2.2 (Login)
Priority	High

4.14.2 EvaluateVerbalResponse.EvaluateDomainKnowledge

Identifier	FR-4.14.2
Title	EvaluateVerbalResponse.EvaluateDomainKnowledge
Requirement	The system shall check for the domain knowledge from the answers.
Source	Farrukh
Rationale	To help users know about their domain knowledge.
Business Rule	An evaluation should be made in accordance to answers provided.
Dependencies	FR-4.14.1
Priority	Medium

4.14.3 EvaluateVerbalResponse.EvaluateUnderstanding

Identifier	FR-4.14.3
Title	EvaluateVerbalResponse.EvaluateUnderstanding
Requirement	The system shall check for the domain Understanding from the answers provided by user
Source	Farrukh
Rationale	To help users know about their understanding.
Business Rule	An evaluation should be made in accordance to answers provided.
Dependencies	FR-4.14.1
Priority	Medium

4.15 UC-15: Evaluate Non-Verbal Response

The following are the functional requirements for Evaluate Non-Verbal Response Use Case

4.15.1 EvaluateNon-VerbalResponse.EvaluateFillerWords

Identifier	FR-4.15.1
Title	EvaluateNon-VerbalResponse.EvaluateFillerWords
Requirement	The system shall analyze the user's audio responses to calculate the average speech rate (words per minute) and detect the frequency of filler words (e.g., "um," "uh"), providing feedback on clarity and articulation.
Source	Farrukh
Rationale	To ensure that users maintain an optimal speaking pace and minimize filler words, enhancing overall communication clarity.
Business Rule	The platform must support the conduct of behavioral interviews.
Dependencies	FR-4.2.2
Priority	High

4.15.2 EvaluateNon-VerbalResponse.EvaluateTone&Pitch

Identifier	FR-4.15.2
Title	EvaluateNon-VerbalResponse.Tone&PitchAnalysis
Requirement	The system shall analyze the tone and pitch of the user's audio responses to assess confidence and engagement.
Source	Farrukh
Rationale	To help users adjust their vocal delivery and improve overall communication.
Business Rule	Analysis should be performed in real time during the interview.
Dependencies	FR-4.15.1
Priority	Medium

4.15.3 EvaluateNon-VerbalResponse.ResponseTime

Identifier	FR-4.15.3
Title	EvaluateNon-VerbalResponse.EvaluateResponseTime
Requirement	The system should check for long pauses or evaluate the time, user is taking to answer a specific question.
Source	Farrukh
Rationale	To offer users details regarding their response time.
Business Rule	Analysis should be made with high accuracy.
Dependencies	FR-4.15.1
Priority	High

4.16 UC-16: View Verbal/Non-Verbal Report

The following are the functional requirements for View Verbal/Non-Verbal Use Case

4.16.1 View Verbal/Non-VerbalReport.GenerateVerbalFeedback

Identifier	FR-4.16.1
Title	ViewVerbal/Non-VerbalReport.GenerateVerbalFeedback
Requirement	The system shall generate a comprehensive verbal feedback summary that includes metrics for answers correctness, understanding etc.
Source	Farrukh
Rationale	To offer users detailed insights for their delivered answers.
Business Rule	The performance report must clearly display verbal feedback with standardized scores and graphical insights.
Dependencies	FR-4.14.1
Priority	High

4.16.2 ViewVerbal/Non-VerbalReport.GenerateNon-VerbalFeedback

Identifier	FR-4.16.2
Title	ViewVerbal/Non-VerbalReport.GenerateNon-VerbalFeedbackSummary
Requirement	The system shall generate a comprehensive verbal feedback summary that includes metrics for answers correctness, understanding etc.
Source	Farrukh
Rationale	To offer users detailed insights for their delivered answers.
Business Rule	The performance report must clearly display verbal feedback with standardized scores and graphical insights.
Dependencies	FR-4.15.1
Priority	High

4.16.3 ViewVerbal/Non-Verbal Report. Verbal/Non-VerbalFeedback

Identifier	FR-4.16.3
Title	ViewVerbal/Non-VerbalReport.Verbal/Non-VerbalFeedback.
Requirement	The system shall allow users to interactively explore detailed breakdowns of both verbal and non-verbal feedback by clicking on individual metrics.
Source	Farrukh
Rationale	To enhance user understanding by providing insights into their performance.
Business Rule	The feedback interface must support interactive exploration of detailed data.
Dependencies	FR-4.16.1, FR-4.16.2
Priority	Medium

4.17 UC-17: Chatbot for Report-Based Querying

The following are the functional requirements of Chatbot for Report-Based Querying.

4.17.1 Chatbot.Interaction

Identifier	FR-4.17.1
Title	Chatbot.Interaction
Requirement	The system shall provide an AI-powered chatbot that allows users to initiate a conversation about their interview performance immediately after a session.
Source	Raees
Rationale	To offer immediate, accessible support and guidance post-interview.
Business Rule	The chatbot must be accessible from the performance report page.
Dependencies	FR-4.16.1, FR-4.16.2
Priority	High

4.17.2 Chatbot.ContextAwareResponse

Identifier	FR-4.17.2
Title	Chatbot.Context-AwareChatbotResponse
Requirement	The system shall analyze the user's interview data and session context to provide personalized, context-aware responses through the chatbot.
Source	Raees
Rationale	To ensure feedback and guidance are tailored to the user's performance.
Business Rule	Chatbot responses must adapt dynamically based on performance metrics.
Dependencies	FR-4.17.1, FR-4.14.1, FR-4.15.1
Priority	High

4.17.3 Chatbot.ResourceRecommendation

Identifier	FR-4.17.3
Title	Chatbot.ResourceRecommendation
Requirement	The system shall allow the chatbot to recommend additional improvement resources such as articles, videos, and tips.
Source	Raees
Rationale	To provide users with resources for further learning and improvement.
Business Rule	Resource recommendations must be current, relevant, and regularly updated.
Dependencies	FR-4.17.1, FR-4.16.1, FR-4.16.2
Priority	Medium

4.17.4 Chatbot.ResponseDisplay

Identifier	FR-4.17.4
Title	Chatbot.ResponseDisplay
Requirement	The system shall display Chatbot response in the chat window in a readable format.
Source	Raees Zakir
Rationale	Users should receive clear and well-structured responses from.
Business Rule	None
Dependencies	FR-4.17.1
Priority	High

4.18 UC-18: Chatbot for Resume-Based Querying

The following are the functional requirements of Chatbot for Resume-Based Querying

4.18.1 Chatbot.Query on General Resume Information

Identifier	FR-4.18.1
Title	Chatbot Basic Interaction
Requirement	The system shall allow users to ask the chatbot questions related to their resume, such as education, work experience, and skills.
Source	Raees
Rationale	To provide users with a way to retrieve and clarify details from their resume.
Business Rule	The chatbot must extract and present accurate details based on the latest uploaded resume.
Dependencies	FR-4.12.2 (Resume Upload and Parsing)
Priority	High

4.18.2 Chatbot.Query on Project-Specific Details

Identifier	FR-4.18.2
Title	Chatbot.QueryonProject-SpecificDetails
Requirement	The system shall allow users to ask the chatbot about specific projects mentioned in their resume.
Source	Raees
Rationale	To enable users to recall and refine their project-related responses.
Business Rule	The chatbot must accurately extract project details.
Dependencies	FR-4.12.2 (Resume Upload and Parsing)
Priority	High

4.18.3 Chatbot.QueryonSkill-BasedInsights

Identifier	FR-4.18.3
Title	Chatbot.QueryonSkill-BasedInsights
Requirement	The system shall allow users to ask the chatbot about their skills listed in the resume, including experience levels and related projects.
Source	Raees
Rationale	To help users identify strengths, weaknesses based on their resume's skillset.
Business Rule	The chatbot must generate responses that accurately link skills with relevant experience from the resume.
Dependencies	FR-4.12.2 (Resume Upload and Parsing)
Priority	High

4.18.4 Chatbot.ResumeOptimizationSuggestions

Identifier	FR-4.18.4
Title	Chatbot.ResumeOptimizationSuggestions.
Requirement	The chatbot shall provide real-time suggestions to improve the resume based on user queries (e.g., "Add metrics to Project X description" or "Include keyword 'Agile' for ATS compatibility").
Source	Raees
Rationale	Users often lack awareness of resume best practices for Applicant Tracking Systems (ATS).
Business Rule	Suggestions must reference specific resume sections and include examples.
Dependencies	FR-4.12.2 (Resume Upload and Parsing)
Priority	Medium

4.19 UC-19: View Overall Feedback

The following are the functional requirements of View Detailed Feedback Use Case

4.19.1 ViewOverallFeedback.GenerateComprehensivePerformanceReport

Identifier	FR-4.19.1
Title	ViewOverallFeedback.GenerateComprehensivePerformanceReport
Requirement	The system shall generate a detailed report that combines both verbal and non-verbal feedback, providing a evaluation of the user's interview performance.
Source	Farrukh
Rationale	To give users a complete understanding of their strengths and areas for improvement by integrating verbal and non-verbal aspects.
Business Rule	The report must present a unified performance summary, including an overall score and segmented analysis.
Dependencies	FR-4.16.1, FR-4.16.2
Priority	High

4.19.2 ViewOverallFeedback.CorrelationBetweenVerbalandNon-Verbal Performance

Identifier	FR-4.19.2
Title	ViewOverallFeedback.CorrelationBetweenVerbalandNon-Verbal Performance
Requirement	The system shall analyze and display relationships between verbal communication (clarity, correctness, confidence) and non-verbal cues.
Source	Farrukh
Rationale	To provide insights into how verbal and non-verbal behaviors influence overall interview performance.
Business Rule	The report must highlight inconsistencies, such as confident speech but poor body language, and suggest improvements.
Dependencies	FR-4.19.1
Priority	High

4.19.3 ViewOverallFeedback.CompositeScoreCalculation

Identifier	FR-4.19.3
Title	ViewOverallFeedback.CompositeScoreCalculation
Requirement	The system shall calculate an overall performance score using weighted averages of verbal (70%) and non-verbal (30%) metrics, adjustable based on job role (e.g., sales roles weight non-verbal higher).
Source	Farrukh
Rationale	Users benefit from a single benchmark to track progress.
Business Rule	Weight adjustments must follow industry standards (e.g., technical roles: verbal 80%, non-verbal 20%).
Dependencies	FR-4.19.1
Priority	High

4.19.4 ViewOverallFeedback.VisualizeFeedbackMetrics

Identifier	FR-4.19.4
Title	ViewOverallFeedback.VisualizeFeedbackMetrics
Requirement	The system shall use graphical representations to display key speech and response quality metrics.
Source	Farrukh
Rationale	Enhance user understanding by visualizing complex data for quick identification of strengths and weaknesses.
Business Rule	Graphs and charts reflect the most recent evaluation data.
Dependencies	FR-4.19.1
Priority	High

4.19.5 ViewOverallFeedback.ProvideActionableImprovementTips

Identifier	FR-4.19.5
Title	ViewOverallFeedback.ProvideActionableImprovementTips
Requirement	The system shall provide targeted, actionable recommendations.
Source	Farrukh
Rationale	Help users identify improvement areas and receive practical guidance.
Business Rule	Improvement tips shall be context-specific, data-driven.
Dependencies	FR-4.19.1
Priority	High

4.20 UC-20: Download Evaluation Report

The following are the functional requirements for Download Evaluation Report Use Case

4.20.1 DownloadEvaluationReport.DownloadReport

Identifier	FR-4.20.1
Title	DownloadEvaluationReport.DownlaodReport
Requirement	The report shall include verbal scores, non-verbal metrics, summary insights, and actionable tips.
Source	Farrukh
Rationale	Users need a complete record of their performance for review and improvement.
Business Rule	All sections must be clearly labeled and organized hierarchically.
Dependencies	FR-4.19.1
Priority	High

4.20.2 Download Evaluation Report.Multi-Format Export Support

Identifier	FR-4.20.2
Title	DownloadEvaluationReport.MultiFormatExportSupport
Requirement	The system shall allow users to download the evaluation report in multiple formats (PDF, DOCX, CSV).
Source	Farrukh
Rationale	Users may need different formats for sharing, printing, or data analysis.
Business Rule	PDF must retain visual formatting (charts, tables), CSV must include raw metrics.
Dependencies	FR-4.20.1
Priority	High

4.21 UC-21: Track Performance

The following are the functional requirements for Track Performance Use Case

4.21.1 TrackPerformance.ProgressDataStorage&Retrieval

Identifier	FR-4.21.1
Title	TrackPerformance.ProgressDataStorage&Retrieval
Requirement	The system shall store all performance metrics from each interview session—including scores, timestamps, and detailed feedback—to allow users to review their historical performance.
Source	Raees
Rationale	To enable users to track their improvement over time and identify performance trends.
Business Rule	All stored data must be securely maintained and retrievable on demand.
Dependencies	FR-4.19.1
Priority	High

4.21.2 TrackPerformance.TrackPerformanceOverTime

Identifier	FR-4.21.2
Title	TrackPerformance.TrackPerformanceOverTime
Requirement	The system shall provide users with a historical performance tracker that visualizes progress across multiple interviews.
Source	Raees
Rationale	Users need to monitor long-term improvements and identify recurring weak areas.
Business Rule	The performance tracker must store and compare past interview data, showing improvements, stagnation, or declines in specific metrics.
Dependencies	FR-4.21.1
Priority	Medium

4.21.3 TrackPerformance.GraphicalVisualizationofTrends

Identifier	FR-4.21.3
Title	TrackPerformance.GraphicalVisualizationofTrends
Requirement	The system shall generate interactive charts and graphs that visually represent performance trends across multiple sessions, covering both verbal and non-verbal metrics.
Source	Raees
Rationale	To provide users with an intuitive visual representation of their progress over time.
Business Rule	Graphical displays must update in real time with each new session's data.
Dependencies	FR-4.21.1
Priority	High

4.21.4 TrackPerformance.GoalSetting&AnalyticsDashboard

Identifier	FR-4.21.4
Title	TrackPerformance.GoalSetting&AnalyticsDashboard
Requirement	The system shall allow users to set personalized performance goals and display a dashboard that tracks their progress.
Source	Raees
Rationale	To motivate users by providing clear targets and detailed insights.
Business Rule	Dashboard metrics must update in real time and accurately.
Dependencies	FR-4.21.1
Priority	Medium

5. Non-Functional Requirements

5.1 Reliability

REL-1: The system shall maintain 99% uptime, ensuring continuous availability for users.

REL-2: The application shall automatically detect and recover from failures within 10 seconds without user intervention.

REL-3: The system shall provide data redundancy and backup mechanisms to prevent data loss in case of unexpected crashes.

REL-4: All user progress and health data shall be auto-saved every 5 minutes to minimize data loss.

REL-5: The system shall operate reliably even under high user loads (e.g., 10,000 concurrent users) without performance degradation.

5.2 Usability

USE-1: The app shall have a simple and easy-to-navigate interface for all users.

USE-2: All key functions shall be accessible within 5 clicks or less from the main dashboard.

USE-3: The chatbot shall respond to user queries within 5 seconds.

USE-4: The system shall ensure that users can complete account registration within 2 minutes to improve onboarding experience.

5.3 Performance

PER-1: The system shall load the main dashboard within 3 seconds under normal network conditions.

PER-2: The app shall process and display health metrics updates within 3 seconds after syncing data.

PER-3: The chatbot shall respond to user queries within 5 seconds for a seamless experience.

PER-4: The system shall handle up to 10,000 concurrent users without performance degradation.

PER-5: The app shall perform database queries and retrieve stored data within 2 seconds.

5.4 Security

SEC-1: All sensitive data shall be encrypted to protect user privacy.

SEC-2: Passwords shall be stored securely using hashing.

SEC-3: Users shall be able to download or delete their personal data anytime.

SEC-4: Users shall get alerts for suspicious login attempts.

SEC-5: Only authorized users shall be able to access safety settings and personal information.

6. External Interface Requirements

The external interface requirements for SkillEdge AI, including communication with users and external software elements are described below:

6.1 User Interfaces Requirements

UI-1: The application shall have a clean and user-friendly interface, ensuring easy navigation for all users during the interview process.

UI-2: The result screen shall display key performance metrics (verbal, non-verbal, and overall scores) in a visually appealing format.

UI-3: The application shall maintain a consistent design across all screens, ensuring uniformity in fonts, icons, and spacing to enhance readability and aesthetics.

UI-4: All interactive elements, such as buttons and input fields, shall be large enough for easy tapping on both mobile and desktop devices.

UI-5: The system shall display animation or progress indicators during interview processing, feedback generation, and response analysis to enhance user experience.

UI-6: Each screen shall have a "**Home**" button or navigation shortcut to allow users to return to the main page easily.

UI-7: The chatbot interface shall be interactive, providing quick reply to suggestions and allowing users to ask interview-related queries seamlessly.

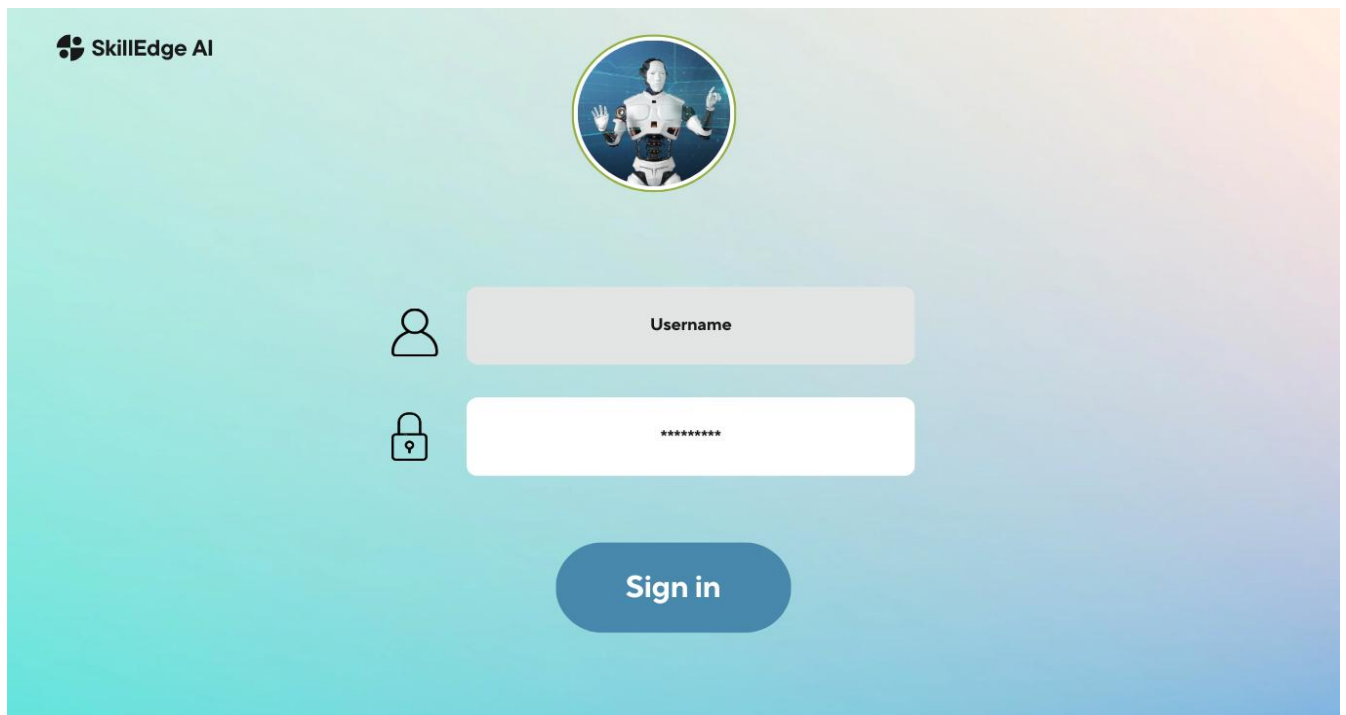
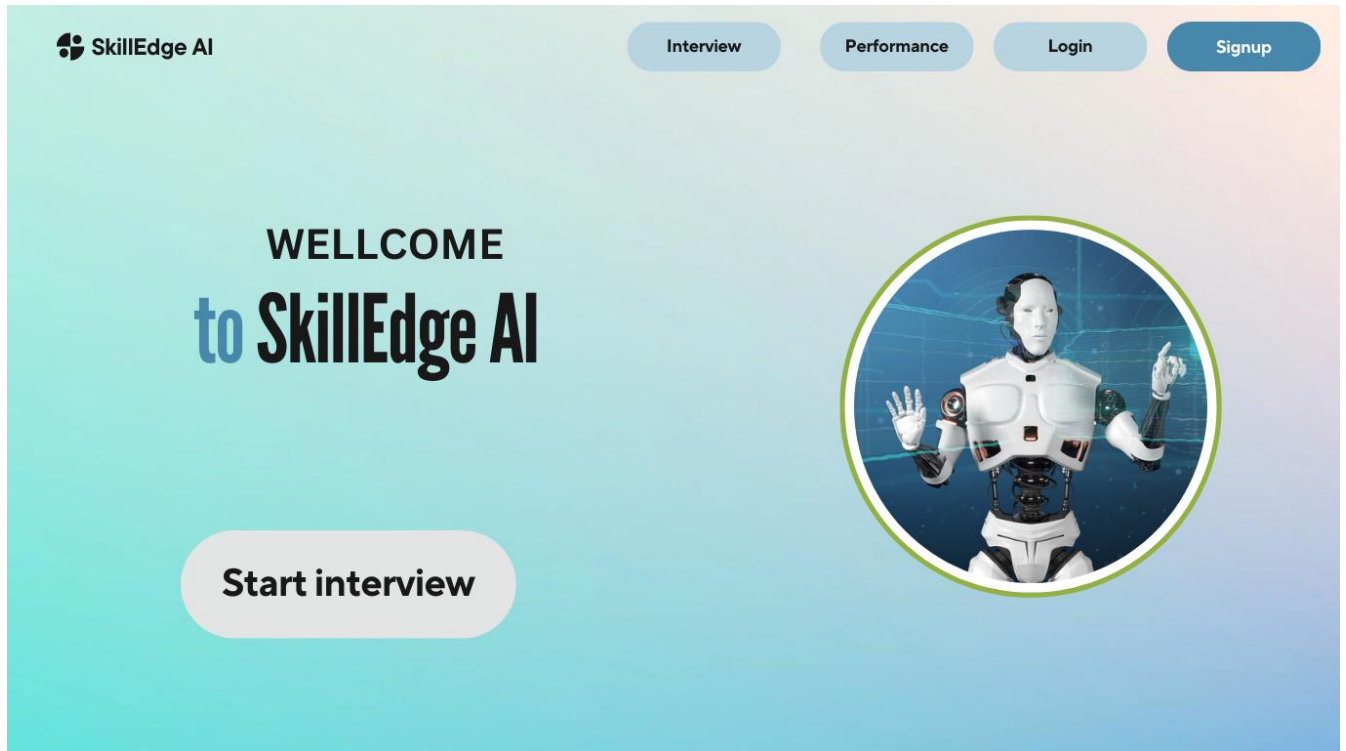
UI-8: The feedback section shall present **verbal and non-verbal insights** using clear **graphs, charts, and color-coded performance indicators** for better visualization.

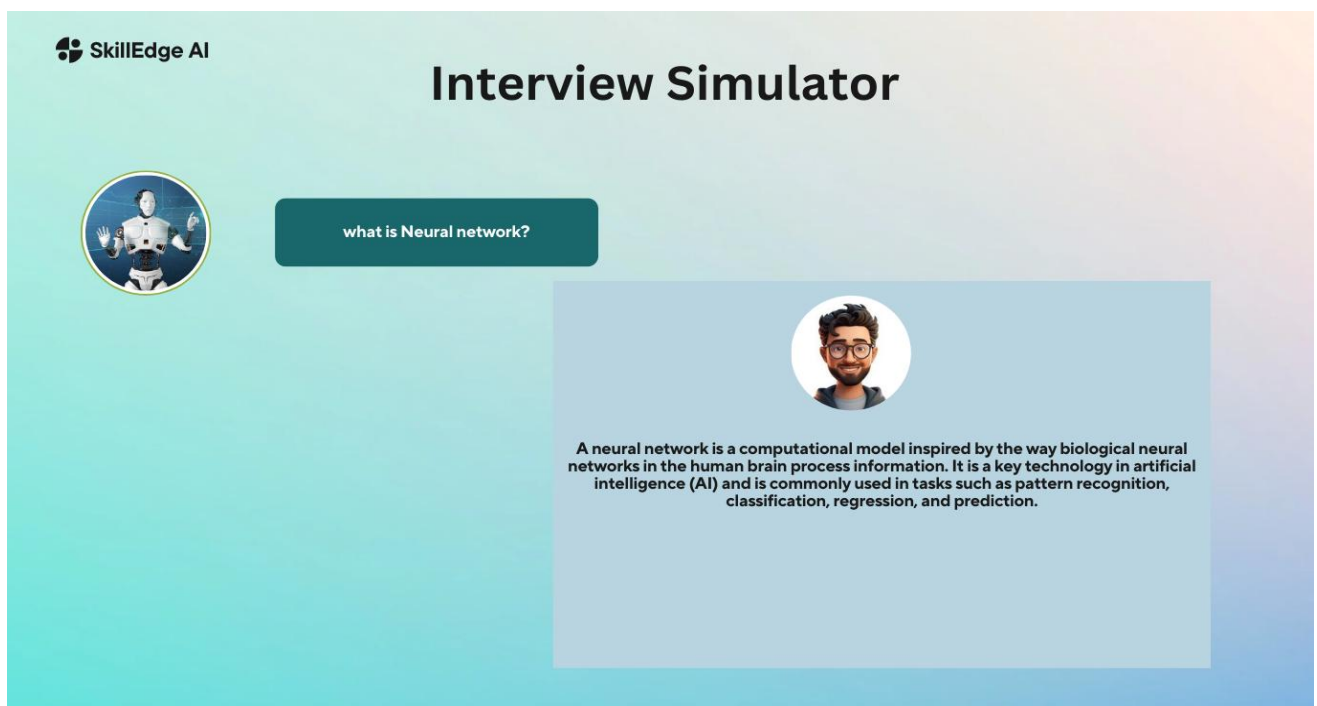
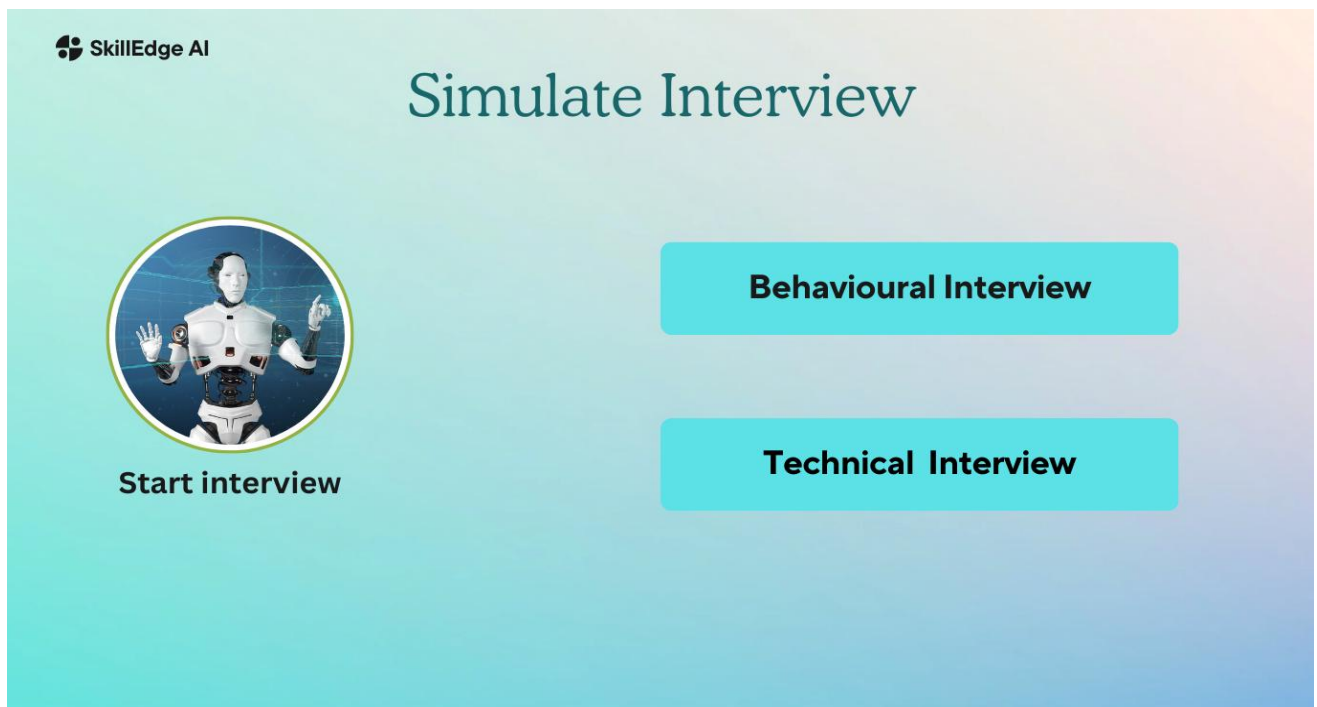
UI-9: The interview simulation interface shall include **real-time visual cues**, such as timers and progress bars, to guide users through the interview.

UI-10: The system shall provide **audio and textual feedback options**, allowing users to receive feedback in multiple formats based on their preference.

6.2 Mockups

Mockups for major modules with description are attached below:







The section includes the SkillEdge AI logo and a large circular avatar of a robot on the left. The title 'Interview Feedback' is prominently displayed. To the right, the heading 'Feedback about interview' is underlined, followed by three paragraphs of feedback text.

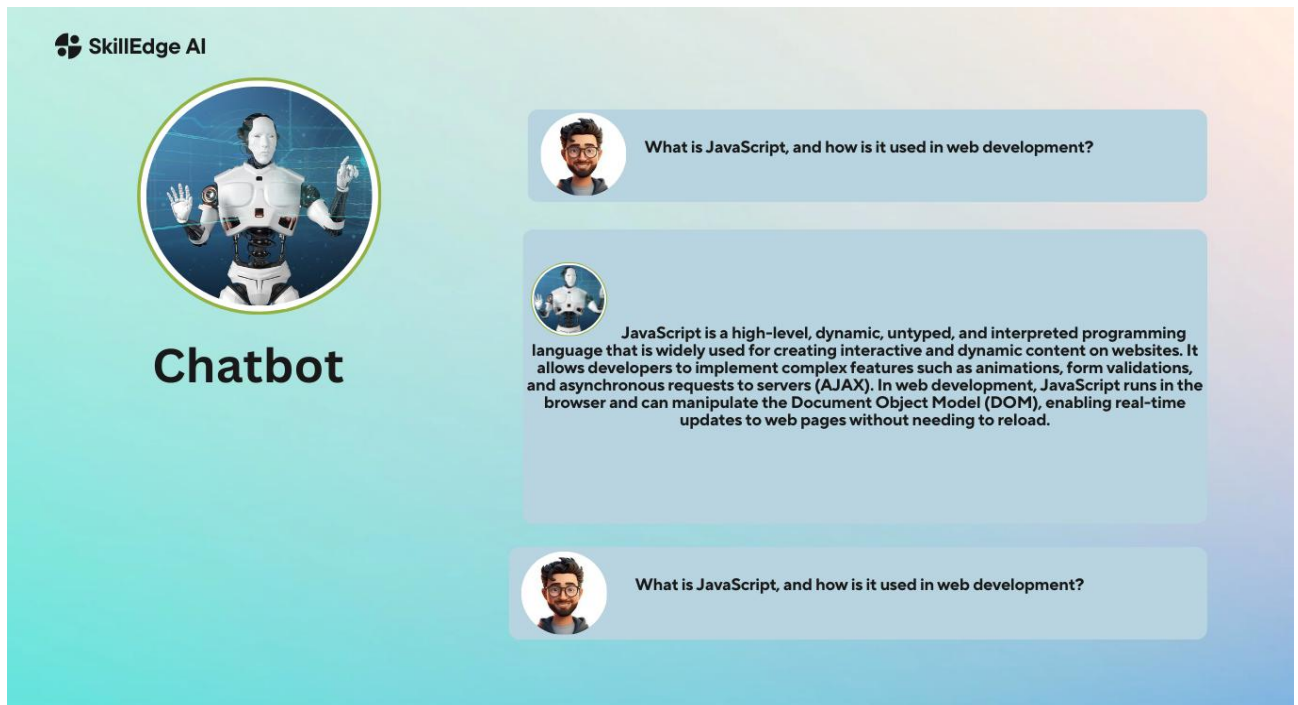
Interview Feedback

Feedback about interview

Depth of Knowledge: Some of the more technical or specific questions revealed gaps in their knowledge, especially in [specific area]. A deeper understanding or experience would be beneficial for this role.

Experience: The candidate's past experience doesn't fully align with the specific requirements of the position. While they have relevant experience, they might need additional training or time to adapt to our systems and processes.

Behavioral Questions: In response to some behavioral questions, the candidate struggled to provide concrete examples. It would help if they prepared more real-life instances to illustrate their skills and approach.



6.3 Software interfaces

SI-1: SkillEdge AI shall be accessible on web-enabled devices that support internet connectivity, including smartphones, tablets, and desktops.

SI-2: The software will be optimized for modern web browsers and will offer mobile support for Android versions 6 and above, and iOS versions 11 and above.

SI-3: The frontend shall be developed using React.js to create a dynamic and user-friendly interface, while seamlessly integrating with the Fast API backend to deliver real-time interactions and efficient data handling.

SI-4: The system shall support secure authentication using Google OAuth as well as traditional email/password methods, enabling users to sign in using their preferred authentication method and ensuring data security during login.

SI-5: The application shall maintain active connections with external services, including MongoDB Atlas for data storage, third-party email services for account verification and notifications, and any required external APIs for supporting machine learning processes and speech-to-text functionalities if needed.

6.4 Hardware interfaces

HI-1: The application shall use the device's microphone to capture the user's audio responses.

HI-2: The application shall use the device's audio output to deliver audio.

HI-3: The application shall require a stable Wi-Fi or mobile network connection.

HI-4: The application shall optionally support the device's webcam for video capture.

6.5 Communications interfaces

CI-1: SkillEdge AI shall utilize Wi-Fi and mobile data for real-time synchronization of interview.

CI-2: The application shall integrate with third-party APIs, such as Google OAuth etc.

CI-3: SkillEdge AI shall support real-time chatbot communication over the internet.

CI-4: The system shall send email notifications for critical updates.

CI-5: All data transmissions between the client and server shall use secure HTTPS protocols.

CI-6: The application shall enable real-time streaming of multimedia content.

7. Conclusion

SkillEdge AI is a cutting-edge, all-in-one interview preparation platform designed to empower job seekers and professionals to excel in competitive interview environments. By integrating advanced AI technologies, the application simulates both technical and behavioral interview scenarios, generating personalized questions and delivering comprehensive, real-time feedback on both verbal content and vocal delivery. With features such as dynamic performance reports, an interactive dashboard displaying detailed metrics, and an intelligent chatbot for report-based guidance, SkillEdge AI provides tailored recommendations that help users refine their communication skills and build confidence. Accessible across multiple platforms, SkillEdge AI is dedicated to continuous improvement, ensuring that every user is equipped with the tools necessary to achieve professional success and secure their desired career opportunities.

8. References

- S. Russell and P. Norvig. *Artificial Intelligence: A Modern Approach*. Upper Saddle River, NJ: Prentice Hall, 2010, pp. 250-300.
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- Lee, S. (2024, February 10). “Interactive Interview Training: Enhancing Communication Skills with AI.” Medium. <https://medium.com/@slee/interactive-interview-training-enhancing-communication-skills-with-ai-abcdef123456>

9. Plagiarism Report

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